

“Katrina’s Children: Evidence on the Structure of Peer Effects from Hurricane Evacuees”

Scott A. Imberman

Adriana D. Kugler

Bruce I. Sacerdote

Web Appendix

1 – Background on Evacuees

1.1 The Hurricane and the Haphazard Evacuation

On August 29, the center of Katrina passed east of New Orleans as a Category 3 hurricane. While the most severe portion of Katrina missed New Orleans, the storm surge caused flooding in 80% of the city.

The extensive flooding stranded many residents. Because of major damage to roads and bridges, there was only one route out of the city. In order to help those still in the city, the Louisiana Superdome was designated as “refuge of last resort” (New York Times, May 22, 2006). By August 30, the number of people sheltering in the Superdome had risen to 20,000 as search and rescue teams kept bringing people from those areas hard-hit by the flooding (Fox News, August 30, 2005). As the Superdome ran out of space, rescuers began dropping off people at the Convention Center, which eventually sheltered another 20,000 people. While the conditions inside the Superdome and the Convention Center were unsanitary and unsafe, the situation outside these shelters was even worse. By August 30, looting was widespread throughout the city and those trying to leave the city without transportation were exposed to the elements and to armed gangs (Fox News, August 30, 2005).

On August 31, Louisiana’s Governor Kathleen Blanco ordered a mandatory evacuation of all of those still remaining in New Orleans and she sent 68 school buses to evacuate those in the Superdome (Times Picayune, Sept. 1, 2005). The evacuation was prolonged as evacuees from across the city started flowing to the Superdome expecting the arena to be the best place to get a ride out of town. By September 3, about 42,000 evacuees had been evacuated from the Superdome and the Convention Center, but a similar number still remained in town. Of those

evacuated, many went to Baton Rouge and other cities in Louisiana. A large number of evacuees also ended up in the Reliant Astrodome in Houston, Texas but by September 1st the Astrodome was full and could not accept any more evacuees. Some ended up in the nearby George R. Brown Convention Center in Houston but others went to large shelters in San Antonio and Dallas, as well as smaller shelters throughout the South. The evacuation process was somewhat chaotic and the destination of evacuees was largely beyond their control. Later, on September 2, the Louis Armstrong International Airport reopened to allow flights for relief efforts and to send evacuees via air to cities willing to help.

1.2 Enrollment of Evacuees

The McKinney-Vento Homeless Education Assistance Act (MVHEAA) of 2001 is a federal law that requires States to enroll homeless students. Since families who had lost their homes or could not return to their home due to the hurricanes were considered homeless, the MVHEAA required schools to accommodate all displaced students. Indeed, the Texas Education Agency (TEA) indicated that “a student who is staying at a shelter or who recently began sharing space with relatives qualifies as homeless.” (http://ritter.tea.state.tx.us/hcane/faq_edsvcs.html)

Schools were also ordered to enroll evacuees promptly by state officials. In Louisiana, the state Superintendent gave orders to enroll students right away and to take care of the details later. East Baton Rouge had already been “registering children from area shelters since they arrived” (CNN, Sept. 9, 2005). On August 31, the Commissioner of Education in Texas announced that evacuee children could register “without having to document residency in the district.” The Superintendent of a school district in the Baton Rouge area told us that administrators approached students in shelters for enrollment “because they knew that most of

them had no transportation.” Also, according to this Superintendant, students were placed “purely based on where they were living at the time...including the shelters and motels.” Thus, students staying in shelters and apartment complexes as well as those staying with family and friends enrolled in the zoned schools. Officials in Houston ISD and East Baton Rouge confirmed that school assignments were done in this way for them as well.

According to an HISD news release, by September 5th 900 evacuees living in shelters around the city had enrolled in school and on September 7th the district began enrolling students at the Reliant complex and the convention center. An HISD official heavily involved in enrollment of evacuees explained to us that district administrators assigned students from the Reliant and Brown shelters to schools based on closeness to the shelters and available space.¹ The registration process was not particularly organized; students were registered as they were found in the shelters until a school was filled and then the HISD officials would move on to enroll students in the next closest school with availability.

2. Data Appendix

2A. Texas Assessment of Knowledge and Skills (TAKS)

In Houston our achievement data is from the Texas Assessment of Knowledge and Skills, the state criterion-referenced accountability exam. This exam is used both for determination of whether a student is promoted to the next grade and accountability ratings. For grade progression, state law requires that students pass the reading test in 3rd grade as well as both reading and math in 5th and 8th grades to be promoted. HISD also requires students in grades 4, 6,

¹ Middle and high school students in Reliant and the convention center were sent to about 10 different existing schools. Elementary students in these shelters were sent to Douglass and Ryan Elementary, which were temporarily opened to house evacuees, as well as to 5 other elementary schools. Nonetheless, Douglass and Ryan were closed by late October 2005 as evacuees exited Reliant and the convention center.

and 7 to pass math and reading for promotion. Students also need to pass the 11th grade exit-level exams in math, English and language arts, science, and social-studies in order to graduate. In other grades there are no passage requirements. All tested grades (3 – 11) count towards school accountability ratings. We consider the 11th grade ELA exam as “reading” for our analyses. In addition, while evacuees had to take the TAKS exams in 2005-06, their performance did not count towards the school’s accountability rating until 2006-07. Nonetheless, one principal we spoke with noted that, while the evacuee scores did not count, the schools still put substantial effort into bringing the evacuees up to proficiency in anticipation of their scores counting the following year.

2B. Louisiana Assessment of Educational Progress (LEAP/iLEAP)

Our main outcome measures are test results for math and English & language arts (ELA) which combines reading and language. Under Louisiana’s accountability program, students in grades 4, 8, and 10 were tested in March of each year prior to 2005-06. These tests are known as the LEAP or Louisiana Educational Assessment Program (grades 4 and 8) and the GEE or Graduation Exit Examination (grade 10). The LEAP are high stakes tests which must be passed to be promoted to the next grade. Students need to pass GEE to graduate. Science and social studies are also tested, but, as is common in the literature, we only consider math and reading. High stakes testing policies were suspended for all 4th and 8th grade students during the 2005-06 school year due to the hurricanes. Additionally, all schools in LA were exempt from accountability requirements during that year. In 2005-06, in response to the No Child Left Behind Act of 2001, Louisiana expanded the testing regime to include grades 3, 5, 6, 7 and 9 for math and ELA. Grades 3, 5, 6, and 7 were also added for science and social studies. Unlike LEAP these exams are based on the Iowa Test of Basic Skills and with questions added to align

the test to criteria required by state and Federal law. In addition, while the iLEAP contributes to determining whether the school meets “adequate yearly progress” under the NCLB act, it is a “low-stakes” exam for students in that their scores do not affect grade advancement. We include LEAP, iLEAP, and GEE in our analysis.

References

CNN. September 9, 2005. “Baton Rouge Swells with Evacuees,”
<http://www.cnn.com/2005/US/09/09/baton.rouge.impact/index.html>.

Fox News. August 30, 2005. “Governor: Evac Superdome, Rescue Centers,”
<http://www.foxnews.com/story/0,2933,167653,00.html>.

New York Times. May 22, 2006. “New Study of Levees Faults Design and Construction,”
<http://www.nytimes.com/2006/05/22/us/22corps.html>.

Times Picayune. September 1, 2005. “School Buses Used to Evacuate,”
http://www.nola.com/weblog/print.ssf?/mtlogs/nola_Times-Picayune/archives/.

TEA Hurricane Related Information. “FAQ: Education Services for Students Evacuated due to Hurricane Katrina,” http://ritter.tea.state.tx.us/hcane/faq_edsvcs.html.

Appendix Table 1 - Observation Counts in Estimating Samples

A. Louisiana													
	Incumbents				Evacuees				All Students				
	Summary Statistics (No restrictions on pre-existing or current outcomes)	Discipline (Must have current discipline measure)	Math (Must have current & a pre- Katrina math score.)	ELA (Must have current & a pre- Katrina ELA score.)	Summary Statistics (No restrictions on pre-existing or current outcomes)	Discipline (Must have current discipline measure)	Math (Must have current & a pre- Katrina math score.)	ELA (Must have current & a pre- Katrina ELA score.)	Summary Statistics (No restrictions on pre- existing or current outcomes)	Discipline (Must have current discipline measure)	Math (Must have current & a pre-Katrina math score.)	ELA (Must have current & a pre-Katrina ELA score.)	
All	892,907	535,363	823,190	809,622	14,422	14,422	12,107	12,103	907,329	549,785	835,297	821,725	
1999-00	38,864	-	38,851	38,863	-	-	-	-	38,864	-	38,851	38,863	
2000-01	43,075	-	43,059	43,073	-	-	-	-	43,075	-	43,059	43,073	
2001-02	81,727	-	76,226	72,082	-	-	-	-	81,727	-	76,226	72,082	
2002-03	81,047	-	77,088	73,013	-	-	-	-	81,047	-	77,088	73,013	
2003-04	112,831	-	101,045	97,937	-	-	-	-	112,831	-	101,045	97,937	
2004-05	116,431	116,431	108,484	106,228	-	-	-	-	116,431	116,431	108,484	106,228	
2005-06	228,568	228,568	208,721	208,653	8,693	8,693	7,167	7,161	237,261	237,261	215,888	215,814	
2006-07	190,364	190,364	169,716	169,773	5,729	5,729	4,940	4,942	196,093	196,093	174,656	174,715	
Elementary	289,467	81,642	287,181	287,230	1,589	1,589	1,349	1,350	291,056	83,231	288,530	288,580	
1999-00	38,864	-	38,851	38,863	-	-	-	-	38,864	-	38,851	38,863	
2000-01	43,075	-	43,059	43,073	-	-	-	-	43,075	-	43,059	43,073	
2001-02	42,968	-	42,955	42,959	-	-	-	-	42,968	-	42,955	42,959	
2002-03	41,527	-	41,517	41,521	-	-	-	-	41,527	-	41,517	41,521	
2003-04	41,391	-	41,384	41,388	-	-	-	-	41,391	-	41,384	41,388	
2004-05	44,084	44,084	44,076	44,081	-	-	-	-	44,084	44,084	44,076	44,081	
2005-06	37,558	37,558	35,339	35,345	1,589	1,589	1,349	1,350	39,147	39,147	36,688	36,695	
Middle/High	603,440	453,721	536,009	522,392	12,833	12,833	10,758	10,753	609,169	459,450	540,949	527,334	
2001-02	38,759	-	33,271	29,123	-	-	-	-	38,759	-	33,271	29,123	
2002-03	39,520	-	35,571	31,492	-	-	-	-	39,520	-	35,571	31,492	
2003-04	71,440	-	59,661	56,549	-	-	-	-	71,440	-	59,661	56,549	
2004-05	72,347	72,347	64,408	62,147	-	-	-	-	72,347	72,347	64,408	62,147	
2005-06	191,010	191,010	173,382	173,308	7,104	7,104	5,818	5,811	191,010	191,010	173,382	173,308	
2006-07	190,364	190,364	169,716	169,773	5,729	5,729	4,940	4,942	196,093	196,093	174,656	174,715	
B. Houston													
	Incumbents				Evacuees				All Students				
	Summary Statistics (No restrictions on pre-existing or current outcomes)	Discipline /Attendance (Must have current & prior year attendance record)	Math (Must have current, prior year & pre-Katrina math score)	Reading (Must have current, prior year & pre-Katrina reading score)	Summary Statistics (No restrictions on pre-existing or current outcomes)	Discipline /Attendance (Must have current attendance record)	Math (Must have current math score)	Reading (Must have current reading score)	Summary Statistics (No restrictions on pre- existing or current outcomes)	Discipline /Attendance (Evacuees must have current attendance record. Incumbents must have current and prior year	Math (Must have current math score)	Reading (Must have current reading score)	
All	676,008	547,275	339,288	324,731	6,641	6,641	3,061	3,246	682,649	553,916	342,349	327,977	
2003-04	178,974	144,516	91,564	80,149	-	-	-	-	178,974	144,516	91,564	80,149	
2004-05	176,718	146,009	92,749	93,535	-	-	-	-	176,718	146,009	92,749	93,535	
2005-06	171,659	143,035	88,286	86,166	4,986	4,986	2,151	2,307	176,645	148,021	90,437	88,473	
2006-07	148,657	113,715	66,689	64,881	1,655	1,655	910	939	150,312	115,370	67,599	65,820	
Elementary	322,524	223,776	90,267	73,923	2,894	2,894	1,139	1,216	325,418	226,670	91,406	75,139	
2003-04	86,638	62,364	27,684	15,913	-	-	-	-	86,638	62,364	27,684	15,913	
2004-05	84,267	61,596	26,470	27,144	-	-	-	-	84,267	61,596	26,470	27,144	
2005-06	81,334	59,822	25,472	22,734	2,121	2,121	810	868	83,455	61,943	26,282	23,602	
2006-07	70,285	39,994	10,641	8,132	773	773	329	348	71,058	40,767	10,970	8,480	
Middle/High	353,484	323,499	249,021	250,808	3,747	3,747	1,922	2,030	357,231	327,246	250,943	252,838	
2003-04	92,336	82,152	63,880	64,236	-	-	-	-	92,336	82,152	63,880	64,236	
2004-05	92,451	84,413	66,279	66,391	-	-	-	-	92,451	84,413	66,279	66,391	
2005-06	90,325	83,213	62,814	63,432	2,865	2,865	1,341	1,439	93,190	86,078	64,155	64,871	
2006-07	78,372	73,721	56,048	56,749	882	882	581	591	79,254	74,603	56,629	57,340	

Incumbent students are restricted to those used in the peer-effects regressions. Hence the LA sample includes grade 4 from 1999-2000 through 2004-2005, grade 5 in 2005-2006, grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. The sample in LA is also limited to schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. To maintain consistency with later regressions we restrict to students that have a pre-Katrina outcome measure, with the exception of Houston evacuees since we do not have any pre-Katrina information on those students. Note that the first two columns for evacuees have the same observation counts as restricting to evacuees with prior achievement would result in dropping all observations in 2005-06 for Houston. Hence we use all evacuees in the summary statistics.

Appendix Table 2: Characteristics of Evacuees and Incumbent Louisiana & Houston Students - 2005-06
Weighted So That Each School Has Weight of 1

	I. All Schools				II. Schools With Positive Evacuee Share			
	A. Houston		B. Louisiana		A. Houston		B. Louisiana	
	Incumbent	Evacuees	Incumbents	Evacuees	Incumbent	Evacuees	Incumbents	Evacuees
Demographics								
Female	0.480 (0.500)	0.469 (0.499)	0.482 (0.500)	0.497 (0.500)	0.488 (0.500)	0.469 (0.499)	0.488 (0.500)	0.497 (0.500)
White	0.072 (0.259)	0.067 (0.248)	0.494 (0.500)	0.374 (0.484)	0.082 (0.275)	0.067 (0.248)	0.488 (0.500)	0.374 (0.484)
Hispanic	0.558 (0.497)	0.096 (0.294)	0.013 (0.113)	0.014 (0.116)	0.525 (0.499)	0.096 (0.294)	0.013 (0.114)	0.014 (0.116)
African-American	0.344 (0.475)	0.823 (0.382)	0.476 (0.500)	0.589 (0.492)	0.362 (0.481)	0.823 (0.382)	0.483 (0.500)	0.587 (0.492)
Asian	0.015 (0.122)	0.277 (0.448)	0.008 (0.089)	0.014 (0.118)	0.029 (0.169)	0.277 (0.448)	0.009 (0.094)	0.014 (0.118)
Free/Reduced Price Lunch	0.828 (0.378)	0.954 (0.211)	0.606 (0.489)	0.836 (0.370)	0.811 (0.391)	0.954 (0.211)	0.619 (0.486)	0.836 (0.370)
At-Risk	0.703 (0.457)	0.933 (0.249)	- -	- -	0.683 (0.465)	0.933 (0.249)	- -	- -
Fraction Katrina/Rita Evacuee in School	0.024 (0.030)	0.029 (0.030)	0.035 (0.072)	0.045 (0.079)	0.029 (0.030)	0.029 (0.030)	0.045 (0.079)	0.045 (0.079)
Fraction Katrina/Rita Evacuee in Grade	0.025 (0.034)	0.040 (0.037)	0.034 (0.072)	0.058 (0.100)	0.029 (0.036)	0.040 (0.037)	0.044 (0.079)	0.058 (0.100)
Observations	171,659	4,986	228,568	8,693	158,603	4,986	207,384	8,693
Test Scores, Discipline, and Attendance								
Math (Standard Deviations)	-0.568 (1.463)	-1.239 (1.353)	-0.116 (1.017)	-0.327 (1.072)	-0.504 (1.440)	-1.239 (1.353)	-0.094 (1.004)	-0.327 (1.072)
Observations	112,241	2,151	227,912	8,664	104,820	2,151	206,797	8,664
Reading / English & Language Arts (Standard	-0.818 (1.862)	-1.394 (1.751)	-0.118 (1.032)	-0.300 (1.110)	-0.720 (1.815)	-1.394 (1.751)	-0.091 (1.020)	-0.030 (1.110)
Observations	171,659	2,307	227,902	8,671	106,355	2,307	206,780	8,671
Disciplinary Infractions	0.464 (1.431)	0.495 (1.449)	0.385 (0.702)	0.407 (0.715)	0.443 (1.403)	0.495 (1.449)	0.357 (0.661)	0.407 (0.715)
Observations	171,659	4,986	228,568	8,693	158,603	4,986	207,384	8,693
Attendance	94.23 (11.15)	87.08 (15.50)	- -	- -	95.33 (7.61)	87.08 (15.50)	- -	- -
Observations	171,659	4,986	-	-	158,603	4,986	-	-

Standard deviations are provided in parentheses. Incumbent students are restricted to those used in the peer-effects regressions. Hence the LA sample includes grade 4 from 1999-2000 through 2004-2005, grade 5 in 2005-2006, grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. The sample in LA is also limited to schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. To maintain consistency with later regressions we restrict to students that have a pre-Katrina outcome measure, with the exception of Houston evacuees since we do not have any pre-Katrina information on those students. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. TAKS scores are standard deviations of scale scores within grade and year using statewide means and standard deviations. When students have multiple scores for a single subject in a given year we use the lowest score.

Appendix Table 3: Reduced Form Estimates of School Level Evacuee Shares on Achievement - Houston

	All	African-American	Boys	Girls	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	Placebo Test
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
A. Elementary									
Math	-0.47 (0.55)	-1.35 (0.83)	-0.32 (0.64)	-0.61 (0.55)	-0.86 (0.96)	-1.53** (0.71)	-1.54* (0.81)	0.51 (0.56)	-0.31 (0.93)
<i>Observations</i>	90,267	23,966	44,868	45,399	20,290	23,159	23,264	23,554	53,255
Reading	-0.53 (0.58)	-2.36** (1.05)	-1.03 (0.68)	-0.06 (0.61)	-1.12 (1.37)	0.25 (0.79)	-0.08 (0.53)	0.55 (0.86)	-0.87 (1.32)
<i>Observations</i>	73,923	21,582	37,264	36,659	16,740	19,388	18,994	18,801	42,281
B. Middle/High									
Math	0.29 (0.39)	-0.13 (0.55)	0.12 (0.43)	0.46 (0.38)	-0.29 (0.94)	-0.31 (0.42)	0.12 (0.36)	-0.54 (0.37)	-0.01 (0.45)
<i>Observations</i>	249,021	72,314	122,799	126,222	51,422	63,680	65,957	67,962	129,845
Reading	0.58 (0.45)	0.39 (0.54)	0.46 (0.50)	0.67 (0.45)	-0.55 (1.71)	0.02 (0.40)	-0.63 (0.41)	-1.28*** (0.38)	-0.30 (0.47)
<i>Observations</i>	250,808	72,641	123,639	127,169	50,142	65,602	66,976	68,088	130,310

Standard errors are provided in parentheses and clustered by school. Regressions cover 2003-04 - 2006-07 and include student's race, gender, free/reduced price lunch status, grade-by-year fixed effects, school fixed-effects, and lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. TAKS scores are standard deviations of scale scores within grade and year based on statewide means and standard deviations. When students have multiple scores for a single subject in a given year we use the lowest score. Placebo regressions are limited to pre-Katrina years and apply the 2005-06 evacuee shares to 2004-05 observations. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year across the district. Elementary is defined as any student in grades 3 - 5. Middle/High is any student in grade 6 - 11. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 4 - Houston Achievement Specification Tests - Grade Level Evacuee Share, Elementary

Model ↓	Pre- Katrina TAKS Quartile →	A. TAKS Math					B. TAKS Reading				
		All	Bottom Quartile	2nd Quartile	3rd Quartile	4th Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	4th Quartile
(1) IV (Shelter Instrument Only)		0.76 (1.34)	-0.51 (2.24)	-0.36 (1.38)	-1.05 (1.07)	0.33 (0.79)	1.28 (1.35)	0.48 (2.44)	1.75 (1.73)	0.09 (0.76)	0.45 (0.89)
(2) IV (Shelter & 9/13/05 Instruments)		-0.47 (0.76)	-1.09 (1.31)	-1.71 (1.05)	-1.32 (0.96)	0.54 (0.73)	-0.27 (0.83)	-0.57 (1.81)	1.15 (1.07)	0.34 (0.66)	1.19 (0.75)
(3) Student & School Fixed Effects		-0.49 (0.34)	0.02 (0.66)	-0.73 (0.45)	-0.77 (0.62)	0.05 (0.57)	0.32 (0.36)	0.39 (0.76)	-0.08 (0.57)	0.82 (0.53)	0.69 (0.57)
Observations		90,267	20,229	23,183	23,368	23,487	73,923	16,813	19,348	18,977	18,785
(4) Class Level Katrina Share (2003-04 through 2005-06 only)		-0.19 (0.26)	0.02 (0.39)	-0.86*** (0.26)	-0.84** (0.37)	-0.14 (0.30)	0.50* (0.26)	0.04 (0.51)	0.61* (0.33)	0.39 (0.34)	0.12 (0.38)
Observations		79,626	17,628	20,524	20,700	20,774	65,791	14,860	17,309	16,915	16,707
(5) Limited to Schools with < 10% Katrina in 0506		-0.19 (0.71)	-1.32 (1.23)	-1.36* (0.81)	-0.72 (0.77)	0.84 (0.84)	-0.38 (0.65)	-0.76 (1.32)	-0.17 (0.88)	-0.02 (0.60)	-0.37 (0.94)
Observations		88,722	19,832	22,761	22,981	23,148	72,617	16,430	18,986	18,682	18,519
(6) Interrupted Panel - 2003-04, 05-06, and 06-07 Only		-0.20 (0.53)	-0.41 (0.97)	-1.04 (0.66)	-0.83 (0.62)	1.05** (0.53)	-0.16 (0.61)	-0.16 (1.12)	0.13 (0.95)	-0.55 (0.55)	0.47 (0.66)
Observations		63,797	14,609	16,307	16,397	16,484	46,779	10,969	12,235	11,878	11,697
Instrument for Evac Share w/ Evac Share in 0405 School											
(7) OLS		-0.37 (0.45)	-0.88 (0.91)	-1.15** (0.54)	-1.25** (0.62)	0.69 (0.53)	-0.33 (0.43)	-0.31 (0.91)	-0.05 (0.65)	-0.08 (0.42)	0.04 (0.58)
Second Stage		-0.23 (0.48)	-0.71 (1.15)	-1.14** (0.57)	-1.12* (0.63)	0.60 (0.53)	-0.41 (0.48)	-0.78 (1.36)	-0.06 (0.71)	-0.19 (0.50)	0.20 (0.58)
Observations		78,995	17,445	20,357	20,531	20,662	65,179	14,679	17,144	16,771	16,585
(8) Unstandardized Scale Scores		-134 (123)	-187 (215)	-297* (152)	-321* (175)	109 (123)	-124 (122)	-201 (260)	-77 (138)	-8 (98)	93 (145)
Observations		90,292	20,297	23,166	23,269	23,560	73,938	16,753	19,389	18,994	18,802
(9) Replace Students Enrolled But Missing Score with Lagged Score		-0.22 (0.39)	-0.60 (0.76)	-1.14** (0.49)	-0.91 (0.56)	0.80 (0.49)	-0.13 (0.39)	-0.56 (0.91)	0.06 (0.55)	0.11 (0.37)	0.23 (0.55)
Observations		92,167	21,306	23,417	23,636	23,808	75,893	17,898	19,540	19,299	19,156
(10) Replace Students Enrolled But Missing Score with Mean Score for Bottom Quartile Students		-0.41 (0.41)	-0.98 (0.78)	-1.16** (0.50)	-0.96 (0.63)	0.24 (0.47)	-0.04 (0.39)	-0.21 (1.00)	0.05 (0.56)	-0.18 (0.38)	0.26 (0.56)
Observations		92,250	21,355	23,434	23,646	23,815	76,036	18,006	19,560	19,307	19,163
Standardize Using District Means and Standard Deviations for 2004-05 Cohort											
(11) Grade Level Katrina Share		-0.15 (0.23)	-0.20 (0.47)	-0.67** (0.29)	-0.49 (0.30)	0.32 (0.28)	-0.01 (0.21)	0.15 (0.43)	-0.00 (0.26)	0.05 (0.18)	0.04 (0.28)
Observations		90,267	20,301	23,143	23,277	23,546	73,923	16,740	19,388	18,990	18,805
(12) School - Year Fixed Effects (2006, 2007 Only)		-0.38 (0.82)	0.06 (1.34)	-0.82 (1.03)	-0.54 (0.98)	1.77 (1.52)	0.88 (1.04)	2.67 (1.88)	1.19 (1.17)	0.87 (1.27)	-1.17 (1.25)
Observations		36,113	8,788	9,048	9,082	9,195	30,866	7,294	7,783	7,851	7,938
(13) School - Grade Fixed Effects		-0.15 (0.47)	-0.56 (0.89)	-1.06* (0.55)	-0.94 (0.62)	0.46 (0.47)	-0.11 (0.43)	-0.37 (1.02)	0.10 (0.67)	-0.01 (0.40)	0.21 (0.55)
Observations		90,267	20,229	23,183	23,368	23,487	73,923	16,813	19,348	18,977	18,785
(14) Evacuee Counts in Grade with Quartic Control in Grade-Level Enrollment		-0.0020 (0.0038)	-0.0069 (0.0080)	-0.0069* (0.0041)	-0.0065 (0.0047)	0.0058 (0.0043)	-0.0004 (0.0035)	-0.0007 (0.0075)	0.0024 (0.0047)	0.0034 (0.0037)	0.0038 (0.0039)
Observations		90,267	20,229	23,183	23,368	23,487	73,923	16,813	19,348	18,977	18,785
Separate Estimates for New Entrants to School											
(15) Evacuee Share in Grade		-0.21 (0.42)	-1.22 (0.97)	-0.99** (0.50)	-0.73 (0.55)	0.93* (0.50)	-0.03 (0.41)	-0.20 (1.10)	0.27 (0.59)	0.21 (0.41)	0.39 (0.55)
Evacuee Share in Grade X New Entrant		-0.37 (0.55)	1.93 (1.23)	-0.94 (0.70)	-1.89* (1.00)	-3.40** (1.58)	-0.59 (0.52)	-1.00 (1.34)	-0.88 (0.91)	-0.42 (0.90)	-2.46 (1.58)
Observations		90,267	20,229	23,183	23,368	23,487	73,923	16,813	19,348	18,977	18,785
(16) Exclude Schools With No Evacuees in 2005-06		-0.19 (0.41)	-0.57 (0.87)	-0.92* (0.49)	-0.88 (0.58)	0.67 (0.50)	0.07 (0.41)	-0.40 (0.96)	0.39 (0.58)	0.18 (0.40)	0.22 (0.57)
Observations		79,203	17,575	20,333	20,465	20,830	65,181	14,581	16,756	16,725	17,119

Standard errors are provided in parentheses and clustered by school. Regressions cover 2003-04 - 2006-07 except where noted and include student's race, gender, free/reduced price lunch status, grade-by-year fixed effects, school fixed-effects, and lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. TAKS scores are standard deviations of scale scores within grade and year using statewide means and standard deviations. When students have multiple scores for a single subject in a given year we use the lowest score. Elementary covers students in grades 3 - 5 prior to 2006, grades 4 & 5 in 2006, and grade 5 in 2007. Middle/High is any student in grades 6 - 11. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 5 - Houston Achievement Specification Tests - Grade Level Evacuee Share, Middle/High

Model ↓	Pre- Katrina TAKS Quartile →	A. TAKS Math					B. TAKS Reading				
		All	Bottom Quartile	2nd Quartile	3rd Quartile	4th Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	4th Quartile
(1)	IV (Shelter Instrument Only)	0.39 (0.92)	1.45 (2.17)	-0.19 (0.67)	-0.50 (0.53)	-0.66 (0.53)	1.56* (0.89)	4.34 (3.25)	0.14 (0.86)	0.05 (0.92)	-1.41** (0.67)
(2)	IV (Shelter & 9/13/05 Instruments)	1.15 (0.88)	1.58 (1.63)	0.57 (0.66)	-0.01 (0.48)	-0.64 (0.48)	1.71** (0.70)	2.86 (2.39)	0.81 (0.59)	0.05 (0.69)	-1.35** (0.63)
(3)	Student & School Fixed Effects	0.83** (0.33)	1.34* (0.68)	0.28 (0.36)	0.42 (0.29)	0.55 (0.37)	0.44 (0.44)	0.92 (1.01)	0.19 (0.35)	-0.29 (0.33)	-0.41 (0.47)
	Observations	249,021	51,011	63,829	66,189	67,992	250,808	50,089	65,660	66,940	68,119
(4)	Limited to Schools with < 10% Katrina in 0506	0.31 (0.55)	-0.04 (1.37)	-0.54 (0.64)	-0.09 (0.45)	-0.31 (0.47)	0.13 (0.62)	-1.96 (1.74)	0.27 (0.55)	-0.03 (0.56)	-1.27*** (0.46)
	Observations	234,603	47,279	59,682	62,422	65,220	236,208	46,248	61,559	63,071	65,330
(5)	Interrupted Panel - 2003-04, 05-06, and 06-07 Only	0.22 (0.37)	0.24 (0.87)	-0.25 (0.46)	0.38 (0.31)	0.14 (0.40)	0.36 (0.45)	-0.48 (1.54)	-0.49 (0.35)	-0.05 (0.36)	-0.81** (0.34)
	Observations	182,742	37,302	46,824	48,694	49,922	184,417	36,892	48,153	49,216	50,156
	Instrument for Evac Share w/ Evac Share in 0405 School										
(6)	OLS	0.76** (0.34)	0.69 (0.93)	0.47 (0.39)	0.55* (0.32)	0.25 (0.43)	0.92** (0.40)	0.45 (1.67)	0.32 (0.40)	-0.08 (0.38)	-0.78* (0.40)
	Second Stage	0.88** (0.38)	1.14 (0.95)	0.34 (0.41)	0.53 (0.39)	0.07 (0.48)	1.11** (0.43)	1.43 (1.66)	0.25 (0.43)	-0.20 (0.47)	-1.11** (0.45)
	Observations	172,319	36,020	44,550	45,509	46,240	173,077	35,168	45,733	45,943	46,233
(7)	Unstandardized Scale Scores	72 (77)	-87 (192)	-104 (83)	42 (74)	-139* (79)	86 (80)	-96 (279)	18 (72)	-102 (67)	-210*** (72)
	Observations	249,021	51,422	63,680	65,957	67,962	250,808	50,142	65,602	66,976	68,088
(8)	Replace Students Enrolled But Missing Score with Lagged Score	-0.22 (0.31)	-0.99* (0.57)	-0.43 (0.39)	-0.05 (0.32)	-0.26 (0.36)	0.28 (0.39)	-0.61 (1.20)	0.11 (0.33)	-0.29 (0.32)	-0.94*** (0.31)
	Observations	266,960	60,665	67,995	68,907	69,393	268,357	60,080	68,999	69,543	69,735
(9)	Replace Students Enrolled But Missing Score with Mean Score for Bottom Quartile Students	0.47 (0.37)	0.80 (0.87)	-0.37 (0.36)	-0.10 (0.34)	-0.17 (0.37)	0.88* (0.50)	0.99 (1.54)	0.18 (0.34)	-0.46 (0.39)	-0.94** (0.39)
	Observations	270,858	63,938	68,364	69,104	69,452	272,559	63,788	69,250	69,706	69,815
	Standardize Using Means and Standard Deviations for										
(10)	2004-05 Cohort	0.21 (0.25)	0.17 (0.57)	-0.26 (0.26)	0.074 (0.20)	-0.25 (0.27)	0.35 (0.26)	0.23 (0.90)	0.09 (0.21)	-0.20 (0.21)	-0.62*** (0.21)
	Observations	249,021	51,406	63,969	65,944	67,975	250,808	50,161	65,573	67,003	68,071
(11)	School - Year Fixed Effects (2006, 2007 Only)	0.31 (0.62)	2.04 (1.36)	-0.04 (1.00)	-0.43 (0.69)	1.15 (1.31)	0.73 (0.71)	1.50 (2.03)	1.15 (0.92)	1.45** (0.56)	0.15 (1.34)
	Observations	118,862	24,079	30,301	31,705	32,777	120,181	23,192	31,281	32,052	32,936
(12)	School - Grade Fixed Effects	0.33 (0.34)	-0.13 (0.84)	-0.35 (0.39)	0.02 (0.32)	-0.29 (0.37)	0.69 (0.42)	0.06 (1.64)	0.11 (0.36)	-0.38 (0.36)	-1.18*** (0.32)
	Observations	249,021	51,011	63,829	66,189	67,992	250,808	50,089	65,660	66,940	68,119
(13)	Evacuee Counts in Grade with Quartic Control in Grade-Level Enrollment	0.0002 (0.0007)	0.0016 (0.0018)	-0.0006 (0.0007)	0.0004 (0.0007)	-0.0003 (0.0005)	0.0006 (0.0011)	0.0009 (0.0035)	-0.0003 (0.0008)	-0.0011 (0.0008)	-0.0015*** (0.0005)
	Observations	249,021	51,011	63,829	66,189	67,992	250,808	50,089	65,660	66,940	68,119
	Separate Estimates for New Entrants to School										
(14)	Evacuee Share in Grade	0.56 (0.37)	0.32 (0.93)	0.00 (0.43)	0.36 (0.41)	0.45 (0.42)	0.75* (0.43)	-0.05 (1.61)	0.18 (0.43)	-0.12 (0.43)	-0.74* (0.43)
	Evacuee Share in Grade X New Entrant	-0.65 (0.59)	-0.27 (1.15)	-0.88* (0.53)	-0.88 (0.69)	-1.96*** (0.72)	-0.26 (0.52)	0.48 (1.25)	0.13 (0.55)	-0.57 (0.47)	-0.87 (0.71)
	Observations	249,021	51,011	63,829	66,189	67,992	250,808	50,089	65,660	66,940	68,119
(15)	Exclude Schools With No Evacuees in 2005-06	0.24 (0.36)	0.01 (0.87)	-0.43 (0.37)	0.002 (0.30)	-0.26 (0.38)	0.63 (0.46)	0.28 (1.65)	0.08 (0.34)	-0.45 (0.35)	-1.16*** (0.33)
	Observations	242,209	48,955	61,891	64,721	66,642	243,901	48,206	63,597	65,150	66,948

Standard errors are provided in parentheses and clustered by school. Regressions cover 2003-04 - 2006-07 except where noted and include student's race, gender, free/reduced price lunch status, grade-by-year fixed effects, school fixed-effects, and lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. TAKS scores are standard deviations of scale scores within grade and year using statewide means and standard deviations. When students have multiple scores for a single subject in a given year we use the lowest score. Elementary covers students in grades 3 - 5 prior to 2006, grades 4 & 5 in 2006, and grade 5 in 2007. Middle/High is any student in grades 6 - 11. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 6: Reduced Form Estimates of Grade Level Evacuee Shares on Achievement by Quintiles - Houston

	Bottom Quintile	2nd Quintile	3rd Quintile	4th Quintile	Top Quintile
	(1)	(2)	(3)	(4)	(5)
A. Elementary					
Math	-0.66 (0.95)	-1.01 (0.62)	-1.22** (0.50)	-0.77 (0.52)	0.64 (0.48)
<i>Observations</i>	15,608	18,566	18,573	18,672	18,848
Reading	-0.63 (1.12)	0.76 (0.55)	0.06 (0.49)	0.16 (0.47)	-0.01 (0.71)
<i>Observations</i>	12,876	15,579	15,350	15,097	15,021
B. Middle/High					
Math	0.05 (1.06)	-0.17 (0.40)	-0.23 (0.33)	-0.01 (0.31)	-0.39 (0.37)
<i>Observations</i>	38,728	50,224	52,163	53,386	54,520
Reading	-0.11 (1.99)	-0.24 (0.34)	0.35 (0.38)	-0.49 (0.40)	-1.23*** (0.41)
<i>Observations</i>	37,291	51,864	53,262	53,762	54,629

Standard errors are provided in parentheses and clustered by school. Regressions cover 2003-04 - 2006-07 and include student's race, gender, free/reduced price lunch status, grade-by-year fixed effects, school fixed-effects, and lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. TAKS scores are standard deviations of scale scores within grade and year based on statewide means and standard deviations. When students have multiple scores for a single subject in a given year we use the lowest score. Placebo regresions are limited to pre-Katrina years and apply the 2005-06 evacuee shares to 2004-05 observations. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year across the district. Elementary is defined as any student in grades 3 - 5. Middle/High is any student in grade 6 - 11. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 7 - The Effect of Disruption from Evacuees on Achievement - Houston

Pre- Katrina TAKS Quartile →	A. Math									
	I. Elementary					II. Middle/High				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	4th Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	4th Quartile
<i>i. OLS w/o Entry</i>										
Avg Weekly Katrina Share in Grade	-0.31 (0.51)	-1.14 (1.08)	-1.37** (0.65)	-1.13 (0.71)	0.97 (0.59)	0.27 (0.51)	-0.14 (1.13)	-0.61 (0.55)	-0.41 (0.42)	-0.51 (0.51)
<i>ii. OLS w/ Entry</i>										
Avg Weekly Katrina Share in Grade	-0.52 (0.55)	-1.04 (1.12)	-1.36** (0.68)	-1.10 (0.73)	0.49 (0.60)	0.30 (0.52)	-0.10 (1.14)	-0.59 (0.55)	-0.39 (0.42)	-0.47 (0.52)
Std Deviation of Weekly Relative Enrollment	0.85 (0.68)	-0.39 (0.68)	-0.03 (0.59)	-0.09 (1.01)	2.05 (1.32)	-0.20** (0.08)	-0.20* (0.11)	-0.10 (0.09)	-0.09 (0.12)	-0.24 (0.27)
<i>iii. 2SLS - First Stage - (SD of Weekly Enrollment is Endogenous)</i>										
Avg Weekly Katrina Share in Grade	0.03 (0.03)	0.03 (0.04)	0.01 (0.03)	0.05 (0.04)	0.04 (0.04)	-0.24 (0.20)	-0.33 (0.34)	-0.27 (0.24)	-0.15 (0.12)	-0.20* (0.12)
Std Deviation of Weekly Katrina Share	0.72*** (0.09)	0.76*** (0.11)	0.80*** (0.09)	0.65*** (0.11)	0.66*** (0.10)	1.70** (0.74)	2.13* (1.21)	1.88** (0.91)	1.30*** (0.37)	1.48*** (0.44)
<i>iv. 2SLS - Second Stage</i>										
Avg Weekly Katrina Share in Grade	-1.21 (0.95)	-2.38 (1.82)	-2.02 (1.26)	-1.10 (1.33)	-0.54 (1.37)	0.05 (0.66)	-0.75 (1.53)	-0.72 (0.64)	-0.95 (0.59)	-0.52 (0.62)
Std Deviation of Weekly Relative Enrollment	3.61 (3.55)	4.72 (5.84)	2.49 (3.81)	-0.11 (4.55)	6.41 (6.05)	1.13 (1.34)	3.04 (3.23)	0.50 (1.16)	3.06 (2.09)	0.06 (1.49)
Observations	90,267	20,229	23,183	23,368	23,487	249,021	51,011	63,829	66,189	67,992
B. Reading										
Pre- Katrina TAKS Quartile →	I. Elementary					II. Middle/High				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	4th Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	4th Quartile
<i>i. OLS w/o Entry</i>										
Avg Weekly Katrina Share in Grade	-0.23 (0.53)	-0.48 (1.26)	0.18 (0.78)	-0.02 (0.49)	-0.04 (0.79)	0.66 (0.63)	-0.23 (2.09)	0.06 (0.50)	-0.58 (0.45)	-1.68*** (0.48)
<i>ii. OLS w/ Entry</i>										
Avg Weekly Katrina Share in Grade	-0.28 (0.54)	-0.50 (1.28)	0.45 (0.81)	0.23 (0.55)	-0.02 (0.83)	0.76 (0.62)	-0.13 (2.09)	0.09 (0.50)	-0.52 (0.45)	-1.52*** (0.48)
Std Deviation of Weekly Relative Enrollment	0.19 (0.32)	0.10 (0.58)	-1.03 (0.69)	-1.02 (1.01)	-0.08 (0.95)	-0.52*** (0.13)	-0.53 (0.37)	-0.15** (0.07)	-0.32*** (0.11)	-1.15*** (0.19)
<i>iii. 2SLS - First Stage - (SD of Weekly Enrollment is Endogenous)</i>										
Avg Weekly Katrina Share in Grade	0.02 (0.03)	-0.01 (0.04)	0.04 (0.04)	0.03 (0.04)	0.03 (0.03)	-0.24 (0.20)	-0.22 (0.24)	-0.30 (0.27)	-0.27 (0.20)	-0.14 (0.09)
Std Deviation of Weekly Katrina Share	0.76*** (0.10)	0.83*** (0.11)	0.70*** (0.12)	0.75*** (0.11)	0.73*** (0.10)	1.68** (0.73)	1.73** (0.83)	2.02* (1.05)	1.73** (0.73)	1.15*** (0.27)
<i>iv. 2SLS - Second Stage</i>										
Avg Weekly Katrina Share in Grade	0.00 (1.15)	0.40 (2.64)	1.45 (1.49)	-0.04 (1.03)	0.96 (1.60)	0.51 (0.79)	-0.51 (2.65)	0.00 (0.58)	-0.62 (0.52)	-1.93*** (0.65)
Std Deviation of Weekly Relative Enrollment	-0.90 (3.83)	-3.42 (7.29)	-4.90 (5.06)	0.05 (3.55)	-3.93 (4.90)	0.81 (1.58)	1.37 (5.06)	0.25 (1.05)	0.25 (1.16)	1.71 (1.90)
Observations	73,923	16,813	19,348	18,977	18,785	250,808	50,089	65,660	66,940	68,119

Standard errors are provided in parentheses and clustered by Grade. Regressions cover 2003-04 - 2006-07 and include student's race, gender, free/reduced price lunch status, grade-by-year fixed effects, Grade fixed-effects, and lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. TAKS scores are standard deviations of scale scores within grade and year using statewide means and standard deviations. When students have multiple scores for a single subject in a given year we use the lowest score. Weekly enrollment and evacuee shares are calculated based on reported dates of entry and exit from the district. Relative enrollment measures use the weekly enrollment counts divided by the October official enrollment count for each Grade. Elementary covers students in grades 3 - 5 prior to 2006, grades 4 & 5 in 2006, and grade 5 in 2007. Middle/High is any student in grades 6 - 11. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 8 - The Effect of Disruption from Evacuees on Achievement - Houston
Early vs. Late Enterers

A. Math										
Pre- Katrina TAKS Quartile →	I. Elementary					II. Middle/High				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	4th Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	4th Quartile
Avg Weekly Katrina Share in Grade	-0.85 (2.34)	-3.23 (3.38)	-0.24 (2.98)	-0.13 (3.08)	-0.08 (3.74)	-0.78 (1.69)	-3.98 (3.88)	-2.09 (2.11)	-5.43*** (1.80)	-0.77 (2.00)
% of Grade Who Are Evacuees Entering in 1st Half of School Year	0.48 (2.00)	1.58 (2.76)	-0.92 (2.44)	-0.63 (2.54)	1.21 (3.24)	0.56 (1.22)	2.93 (3.08)	1.05 (1.74)	4.09*** (1.36)	0.28 (1.49)
% of Grade Who Are Evacuees Entering in 2nd Half of School Year	-1.37 (4.15)	8.72 (8.18)	-0.66 (3.46)	-7.65 (6.84)	-11.29 (8.23)	4.80* (2.78)	5.71 (7.54)	2.98 (3.70)	2.54 (3.05)	-1.03 (4.93)
Observations	90,267	20,229	23,183	23,368	23,487	249,021	51,011	63,829	66,189	67,992
C. Attendance										
Pre- Katrina TAKS Quartile →	I. Elementary					II. Middle/High				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	4th Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	4th Quartile
Avg Weekly Katrina Share in Grade	-2.46 (2.46)	-0.92 (5.35)	0.46 (2.91)	-3.86 (2.62)	-6.07 (3.92)	-1.16 (2.07)	-3.95 (6.23)	-1.82 (1.80)	-1.90 (1.69)	-3.35 (2.17)
% of Grade Who Are Evacuees Entering in 1st Half of School Year	1.92 (1.96)	0.42 (4.05)	-0.11 (2.26)	3.46 (2.15)	5.45* (3.09)	1.28 (1.67)	3.03 (5.19)	1.52 (1.45)	1.33 (1.42)	1.49 (1.68)
% of Grade Who Are Evacuees Entering in 2nd Half of School Year	-2.74 (3.98)	-2.32 (2.59)	-5.99 (5.34)	-11.03 (6.73)	-16.04 (13.70)	3.98 (3.44)	2.94 (9.76)	1.20 (3.30)	-3.11 (3.26)	-1.30 (3.54)
Observations	73,923	16,813	19,348	18,977	18,785	250,808	50,089	65,660	66,940	68,119
D. Discipline										
Pre- Katrina TAKS Quartile →	I. Elementary					II. Middle/High				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	4th Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	4th Quartile
Avg Weekly Katrina Share in Grade	-2.46 (2.46)	-0.92 (5.35)	0.46 (2.91)	-3.86 (2.62)	-6.07 (3.92)	-1.16 (2.07)	-3.95 (6.23)	-1.82 (1.80)	-1.90 (1.69)	-3.35 (2.17)
% of Grade Who Are Evacuees Entering in 1st Half of School Year	1.92 (1.96)	0.42 (4.05)	-0.11 (2.26)	3.46 (2.15)	5.45* (3.09)	1.28 (1.67)	3.03 (5.19)	1.52 (1.45)	1.33 (1.42)	1.49 (1.68)
% of Grade Who Are Evacuees Entering in 2nd Half of School Year	-2.74 (3.98)	-2.32 (2.59)	-5.99 (5.34)	-11.03 (6.73)	-16.04 (13.70)	3.98 (3.44)	2.94 (9.76)	1.20 (3.30)	-3.11 (3.26)	-1.30 (3.54)
Observations	73,923	16,813	19,348	18,977	18,785	250,808	50,089	65,660	66,940	68,119

Standard errors are provided in parentheses and clustered by Grade. Regressions cover 2003-04 - 2006-07 and include student's race, gender, free/reduced price lunch status, grade-by-year fixed effects, Grade fixed-effects, and lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. TAKS scores are standard deviations of scale scores within grade and year using statewide means and standard deviations. When students have multiple scores for a single subject in a given year we use the lowest score. Weekly enrollment and evacuee shares are calculated based on reported dates of entry and exit from the district. Relative enrollment measures use the weekly enrollment counts divided by the October official enrollment count for each Grade. Elementary covers students in grades 3 - 5 prior to 2006, grades 4 & 5 in 2006, and grade 5 in 2007. Middle/High is any student in grades 6 - 11. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 9: Linear-in-Means Models of Peer Effects in Achievement for Houston
OLS and 2SLS Models Using Grade Level Evacuee Share as an Instrument for Peer Achievement - Middle/High

Dependent Variable →	<i>OLS - Pre Katrina Only</i>	<i>OLS - All Years</i>	<i>2SLS - First Stage</i>	<i>2SLS - Second Stage</i>	<i>OLS Model with Peer Achievement Interacted with Own Quartile</i>				<i>Reduced Form with Peer Achievement Interacted with Own Quartile</i>				<i>2SLS Model with Peer Achievement Interacted with Own Quartile - Second Stage Instrumented with Evacuee Share x Own Quartile</i>			
	Avg Peer Score	Avg Peer Score	Evacuee Share	Avg Peer Score	Avg Peer Score x Quartile 1	Avg Peer Score x Quartile 2	Avg Peer Score x Quartile 3	Avg Peer Score x Quartile 4	Evacuee Share x Quartile 1	Evacuee Share x Quartile 2	Evacuee Share x Quartile 3	Evacuee Share x Quartile 4	Avg Peer Score x Quartile 1	Avg Peer Score x Quartile 2	Avg Peer Score x Quartile 3	Avg Peer Score x Quartile 4
	(1)	(2)	(3)	(4)	(5)				(6)				(6)			
Math	0.39*** (0.04)	0.43*** (0.03)	0.36 (0.46)	0.87 (0.72)	0.54*** (0.05)	0.27*** (0.04)	0.24*** (0.04)	0.22*** (0.04)	0.34 (0.87)	-0.30 (0.39)	0.05 (0.30)	-0.27 (0.37)	2.71 (9.87)	-1.16 (3.07)	0.14 (0.82)	-0.58 (1.18)
Observations	130,154	249,012	249,012	249,012	249,012				249,012				249,012			
F - Test of Joint Equality					10.0***				0.4				0.1			
Reading	0.34*** (0.04)	0.38*** (0.04)	-0.05 (0.54)	-11.77 (119.92)	0.61*** (0.05)	0.13*** (0.03)	0.15*** (0.03)	0.19*** (0.05)	0.22 (1.61)	0.22 (0.34)	-0.33 (0.35)	-1.06*** (0.32)	-0.66 (5.52)	-1.68 (9.03)	4.43 (30.28)	-4.47 (12.89)
Observations	130,619	250,797	250,797	250,797	250,797				250,797				250,797			
F - Test of Joint Equality					22.0***				4.0***				0.9			

Standard errors are provided in parentheses and clustered by school. Regressions cover 2003-04 - 2006-07 and include student's race, gender, free/reduced price lunch status, grade-by-year fixed effects, school fixed-effects, and lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. TAKS scores are standard deviations of scale scores within grade and year using statewide means and standard deviations. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year across the district. Elementary is defined as any student in grades 3 - 5. Middle/High is any student in grade 6 - 11. *, **, and *** reflect significance at the 10%, 5%, and 1% levels respectively.

Appendix Table 10: Linear-in-Means Models of Peer Effects in Achievement for Houston
OLS and 2SLS Models Using Grade Level Evacuee Share as an Instrument for Peer Achievement

	OLS Model with Peer Achievement Interacted with Own Quartile without Lags				Reduced Form with Peer Achievement Interacted with Own Quartile				2SLS Model with Peer Achievement Interacted with Own Quartile - Second Stage Instrumented with Evacuee Share x Own Quartile without Lags			
Dependent Variable →	Avg Peer Score x Quartile 1	Avg Peer Score x Quartile 2	Avg Peer Score x Quartile 3	Avg Peer Score x Quartile 4	Evacuee Share x Quartile 1	Evacuee Share x Quartile 2	Evacuee Share x Quartile 3	Evacuee Share x Quartile 4	Avg Peer Score x Quartile 1	Avg Peer Score x Quartile 2	Avg Peer Score x Quartile 3	Avg Peer Score x Quartile 4
	(1)				(2)				(3)			
	I. No Lags											
	A. Elementary											
Math	0.85*** (0.04)	0.77*** (0.05)	0.66*** (0.06)	0.57*** (0.07)	-1.61 (1.12)	-1.11** (0.53)	-1.04* (0.59)	0.61 (0.48)	0.91 (0.59)	0.68** (0.31)	0.66* (0.34)	-0.44 (0.43)
Observations	90,259				90,259				90,259			
F - Test of Joint Equality	7.0***				2.6*				1.4			
Reading	0.88*** (0.05)	0.67*** (0.06)	0.58*** (0.08)	0.37*** (0.07)	-0.90 (1.17)	0.13 (0.58)	0.17 (0.38)	0.36 (0.58)	0.67 (0.78)	-0.15 (0.70)	-0.17 (0.43)	-0.28 (0.53)
Observations	73,918				73,918				73,918			
F - Test of Joint Equality	17.0***				0.5				0.7			
	B. Middle/High											
Math	0.77*** (0.07)	0.27*** (0.04)	0.24*** (0.04)	0.26*** (0.04)	1.14 (1.10)	-0.19 (0.39)	0.17 (0.31)	0.17 (0.35)	8.00 (26.20)	-0.72 (2.41)	0.46 (0.87)	0.37 (0.68)
Observations	249,012				249,012				249,012			
F - Test of Joint Equality	18.3***				1.1				0.1			
Reading	0.86*** (0.08)	0.13*** (0.03)	0.15*** (0.02)	0.21*** (0.05)	0.31 (1.66)	0.37 (0.35)	-0.28 (0.36)	-0.95*** (0.30)	-0.94 (6.20)	-2.9 (14.48)	3.89 (26.80)	-3.93 (11.40)
Observations	250,797				250,797				250,797			
F - Test of Joint Equality	34.2***				4.7***				0.2			
	II. Lags Not Interacted With Quartiles											
	A. Elementary											
Math	0.86*** (0.04)	0.77*** (0.05)	0.67*** (0.05)	0.58*** (0.07)	-1.36 (0.99)	-1.15** (0.51)	-0.99* (0.57)	0.73 (0.50)	0.76 (0.51)	0.70** (0.30)	0.63* (0.32)	-0.52 (0.46)
Observations	90,259				90,259				90,259			
F - Test of Joint Equality	6.2***				3.1**				1.5			
Reading	0.89*** (0.05)	0.67*** (0.06)	0.59*** (0.08)	0.37*** (0.07)	-0.81 (1.05)	0.11 (0.58)	0.14 (0.38)	0.18 (0.57)	0.60 (0.71)	-0.13 (0.70)	-0.14 (0.42)	-0.14 (0.48)
Observations	73,918				73,918				73,918			
F - Test of Joint Equality	15.5***				0.4				0.4			
	B. Middle/High											
Math	0.70*** (0.06)	0.27*** (0.37)	0.24*** (0.04)	0.22*** (0.04)	0.21 (0.84)	-0.33 (0.39)	0.03 (0.30)	-0.36 (0.38)	1.65 (6.49)	-1.25 (3.22)	0.08 (0.82)	-0.77 (1.37)
Observations	249,012				249,012				249,012			
F - Test of Joint Equality	18.8***				0.4				0.1			
Reading	0.76*** (0.07)	0.13*** (0.03)	0.15*** (0.03)	0.19*** (0.05)	-0.36 (1.49)	0.21 (0.34)	-0.34 (0.35)	-1.12*** (0.34)	1.06 (3.64)	-1.57 (8.53)	4.59 (31.47)	-4.70 (13.50)
Observations	250,797				250,797				250,797			
F - Test of Joint Equality	31.0***				4.0***				0.3			

Standard errors are provided in parentheses and clustered by school. Regressions cover 2003-04 - 2006-07 and include student's race, gender, free/reduced price lunch status, grade-by-year fixed effects, school fixed-effects, and lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. TAKS scores are standard deviations of scale scores within grade and year using statewide means and standard deviations. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year across the district. Elementary is defined as any student in grades 3 - 5. Middle/High is any student in grade 6 - 11. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 11: Effects of Evacuees on Test Taking

A. Louisiana - Ratio of Enrollment to Test-Takers in Grade (Middle/High)

	Math	ELA
Katrina/Rita Share in Quartile 1	2.52 (1.80)	2.72 (1.940)
Katrina/Rita Share in Quartile 2	-0.53 (1.550)	-4.46** (2.260)
Katrina/Rita Share in Quartile 3	-6.61** (2.550)	0.26 (3.090)
Katrina/Rita Share in Quartile 4	-2.00 (2.500)	-2.73 (2.490)
Observations (School-Grade-Year)	8,115	8,115

B. Houston - Likelihood of Taking Both Math & Reading Tests Conditional on Enrollment

	Elementary	Middle/High
Katrina/Rita Share	-0.203* (0.122)	-0.374*** (0.116)
Observations (Student-Year)	239,844	401,232

Standard errors are provided in parentheses and clustered by school. Houston: Regressions cover 2003-04 - 2006-07 and include student's race, gender, free/reduced price lunch status, grade-by-year fixed effects, and school fixed-effects. Elementary is defined as any student in grades 3 - 5. Middle/High is any student in grade 6 - 11. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively. Louisiana: Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, and grade-by-year fixed-effects. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 12: Reduced Form Estimates of Evacuee Share of Enrollment in School on Incumbent Test Scores - Louisiana, Elementary (No Lags)

	All	African-American	Boys	Girls	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	Placebo Test
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Math	0.57*** (0.19)	0.52 (0.34)	0.44* (0.24)	0.75*** (0.23)	-0.06 (0.32)	0.01 (0.23)	0.26 (0.26)	0.41 (0.29)	-0.24 (0.23)
<i>Observations</i>	<i>287,181</i>	<i>129,579</i>	<i>147,813</i>	<i>139,368</i>	<i>65,025</i>	<i>71,606</i>	<i>74,138</i>	<i>76,412</i>	<i>200,893</i>
English & Language Arts	-0.01 (0.18)	0.04 (0.31)	0.01 (0.26)	-0.03 (0.19)	-0.50 (0.35)	-0.22 (0.20)	-0.40* (0.23)	-0.23 (0.24)	-0.20* (0.11)
<i>Observations</i>	<i>287,230</i>	<i>129,602</i>	<i>147,842</i>	<i>139,388</i>	<i>65,210</i>	<i>72,013</i>	<i>73,885</i>	<i>76,122</i>	<i>200,930</i>

Standard errors are provided in parentheses and clustered by school. Regressions cover 2003-04 - 2006-07 and include student's race, gender, free/reduced price lunch status, grade-by-year fixed effects, and school fixed-effects. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Placebo regresions are limited to pre-Katrina years and apply the 2006-07 and 2007-08 evacuee shares to 2003-04 and 2004-05 observations, respectively. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006.

Appendix Table 13 Reduced Form Estimates of Evacuee Share of Enrollment in School on Incumbent Test Scores - Louisiana
Middle & High School

	All	African- American	Boys	Girls	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	Placebo Test
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Math	-0.24* (0.14)	-0.39* (0.21)	-0.21 (0.16)	-0.28* (0.15)	-0.38* (0.22)	-0.24 (0.20)	-0.11 (0.14)	-0.12 (0.25)	-0.05 (0.09)
<i>Observations</i>	536,009	229,678	263,450	272,559	123,316	133,757	137,800	141,136	192,548
English & Language Arts	-0.34*** (0.12)	-0.65*** (0.21)	-0.39*** (0.14)	-0.31** (0.13)	-0.30 (0.27)	-0.34** (0.17)	-0.16 (0.17)	-0.13 (0.20)	-0.08 (0.08)
<i>Observations</i>	522,392	218,930	259,140	263,252	116,304	128,894	135,911	141,283	178,958

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Placebo regresions are limited to pre-Katrina years and apply the 2006-07 and 2007-08 evacuee shares to 2003-04 and 2004-05 observations, respectively. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and lagged achievement from the student's most recent pre-Katrina lagged score interacted with the number of years between exams. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007.

*, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 14- Linear-in-Means Models of Peer Effects in Achievement for Louisiana
OLS and 2SLS Models Using Evacuee Share in Grade x Mean Evacuee Lagged Achievement as an Instrument for Peer Achievement - Middle School

	OLS - Pre Katrina Only	OLS - All Years	2SLS - First Stage	2SLS - Second Stage	OLS Model with Peer Achievement Interacted with Own Quartile				Reduced-Form Model with Peer Achievement Interacted with Own Quartile				2SLS Model with Peer Achievement Interacted with Own Quartile - Second Stage; Instrumented with Evac Share x Mean Lagged Evacuee Score x Own Quartile			
	Avg Peer Score	Avg Peer Score	Evac Share x Mean Lagged Evac Score	Avg Peer Score	Avg Peer Score x Quartile 1	Avg Peer Score x Quartile 2	Avg Peer Score x Quartile 3	Avg Peer Score x Quartile 4	Avg Evacuee Score x Evacuee Share x Quartile 1	Avg Evacuee Score x Evacuee Share x Quartile 2	Avg Evacuee Score x Evacuee Share x Quartile 3	Avg Evacuee Score x Evacuee Share x Quartile 4	Avg Peer Score x Quartile 1	Avg Peer Score x Quartile 2	Avg Peer Score x Quartile 3	Avg Peer Score x Quartile 4
	(1)	(2)	(3)	(4)	(5)				(6)				(7)			
A. No Lags																
Math	0.56*** (0.03)	0.52*** (0.07)	2.52*** (0.82)	0.16* (0.09)	0.30*** (0.03)	0.30*** (0.04)	0.27*** (0.09)	0.40*** (0.14)	0.59*** (0.23)	1.08*** (0.21)	0.56*** (0.21)	0.38 (0.29)	0.30*** (0.10)	0.58*** (0.20)	0.17 (0.10)	0.13 (0.14)
Observations	192,843	535,302	535,302	535,302	535,302				535,302				535,302			
F - Test of Joint Equality					4.9***				4.2***				4.6***			
ELA	0.49*** (0.03)	0.61*** (0.02)	1.90*** (0.43)	0.15* (0.09)	0.49*** (0.02)	0.42*** (0.02)	0.32*** (0.02)	0.28*** (0.04)	0.47 (0.32)	0.34 (0.22)	0.87*** (0.20)	0.24 (0.27)	0.30* (0.17)	0.16 (0.11)	0.51*** (0.15)	0.12 (0.17)
Observations	179,217	520,569	520,569	520,569	520,569				520,569				520,569			
F - Test of Joint Equality					23.9***				3.5**				2.9**			
B. Lags Not Interacted With Quartiles																
Math	0.31*** (0.04)	0.34*** (0.06)	2.55*** (0.82)	0.15* (0.08)	0.25*** (0.03)	0.28*** (0.04)	0.27*** (0.09)	0.41*** (0.14)	0.63*** (0.23)	1.32*** (0.22)	0.87*** (0.24)	0.74** (0.30)	0.32*** (0.10)	0.72*** (0.25)	0.26* (0.15)	0.26 (0.19)
Observations	192,843	535,302	535,302	535,302	535,302				535,302				535,302			
F - Test of Joint Equality					2.3*				4.9***				4.9***			
ELA	0.25*** (0.02)	0.41*** (0.02)	1.91*** (0.43)	0.08 (0.08)	0.42*** (0.02)	0.40*** (0.02)	0.33*** (0.02)	0.28*** (0.04)	0.58** (0.27)	0.56** (0.24)	1.15*** (0.22)	0.43 (0.26)	0.37*** (0.13)	0.27** (0.12)	0.67*** (0.20)	0.22 (0.20)
Observations	179,217	520,569	520,569	520,569	520,569				520,569				520,569			
F - Test of Joint Equality					11.2***				4.8***				2.5*			

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Placebo regressions are limited to pre-Katrina years and apply the 2006-07 and 2007-08 evacuee shares to 2003-04 and 2004-05 observations, respectively. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and lagged achievement from the student's most recent pre-Katrina lagged score interacted with the number of years between exams. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 15 - Nonlinear Models of Evacuee Share in School by Evacuee Achievement on Incumbent Achievement - Louisiana

	A. Elementary (No Lags)					B. Middle/High (With Lags)				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Math										
Katrina/Rita Share in Quartile 1	0.51 (0.62)	-1.54** (0.78)	-0.60 (0.61)	-2.00** (0.87)	0.04 (1.20)	-1.04** (0.45)	-1.51** (0.63)	-2.67*** (0.56)	-1.79*** (0.54)	-2.68*** (0.98)
Katrina/Rita Share in Quartile 2	-2.12** (0.89)	-1.35 (1.16)	-1.46 (1.02)	-2.07 (1.32)	-4.70** (2.07)	-1.10* (0.58)	-0.82 (0.77)	-1.32* (0.69)	-1.66** (0.79)	-2.10** (1.05)
Katrina/Rita Share in Quartile 3	1.56 (1.00)	1.43 (1.38)	0.26 (1.04)	0.86 (1.06)	0.55 (1.48)	0.98 (0.66)	0.15 (0.89)	1.35 (0.83)	2.23** (0.87)	1.69 (1.28)
Katrina/Rita Share in Quartile 4	2.37** (1.15)	3.15** (1.60)	2.40** (1.11)	5.01*** (1.40)	5.18*** (1.47)	0.44 (0.51)	2.10** (0.85)	2.29*** (0.69)	0.59 (0.63)	1.77* (1.03)
Observations	287,181	65,025	71,606	74,138	76,412	536,009	123,316	133,757	137,800	141,136
ELA										
Katrina/Rita Share in Quartile 1	0.23 (0.61)	-1.31 (0.96)	-0.28 (0.67)	-0.85 (0.71)	-1.25 (0.85)	-1.46*** (0.51)	-2.81*** (0.73)	-2.63*** (0.64)	-2.99*** (0.56)	-2.82*** (0.76)
Katrina/Rita Share in Quartile 2	-1.08 (0.95)	-1.57 (1.55)	-0.67 (1.04)	-2.50** (1.05)	-0.88 (1.21)	0.23 (0.57)	1.55 (0.98)	-0.29 (0.76)	-0.92 (0.75)	-0.28 (0.99)
Katrina/Rita Share in Quartile 3	-0.04 (0.78)	0.43 (1.38)	-0.22 (0.96)	-0.53 (0.95)	-0.48 (1.05)	-0.68 (0.65)	-0.24 (1.27)	-0.61 (0.86)	0.26 (0.82)	-0.64 (1.02)
Katrina/Rita Share in Quartile 4	0.40 (0.69)	0.62 (1.73)	0.12 (0.83)	1.61* (0.92)	1.44* (0.83)	0.75 (0.48)	2.08* (1.08)	2.66*** (0.70)	2.95*** (0.69)	2.39*** (0.85)
Observations	287,230	65,210	72,013	73,885	76,122	522,392	116,304	128,894	135,911	141,283

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 16- Nonlinear Models of Evacuee Share in School and Evacuee Achievement on Incumbent Achievement - Louisiana
Evacuee Share by Test Administrator

	A. Elementary (No Lags)					B. Middle/High (Lags)				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Math										
Katrina/Rita Share in Quartile 1	-0.67** (0.28)	-0.71** (0.30)	-1.04*** (0.29)	-0.45 (0.29)	-0.65 (0.50)	-0.60*** (0.07)	-0.50*** (0.10)	-0.64*** (0.12)	-0.77*** (0.11)	-1.08*** (0.19)
Katrina/Rita Share in Quartile 2	-0.003 (0.34)	0.04 (0.62)	0.06 (0.32)	0.05 (0.34)	-1.02** (0.48)	-0.09 (0.08)	0.41*** (0.15)	-0.08 (0.12)	-0.11 (0.09)	-0.45*** (0.16)
Katrina/Rita Share in Quartile 3	0.43 (0.37)	0.72 (0.51)	-0.25 (0.34)	-0.05 (0.41)	0.13 (0.57)	0.18* (0.09)	0.88*** (0.16)	0.38*** (0.13)	0.13 (0.10)	-0.15 (0.15)
Katrina/Rita Share in Quartile 4	2.87*** (0.44)	1.10* (0.58)	1.38*** (0.43)	1.93*** (0.40)	2.20*** (0.58)	0.45*** (0.11)	0.60*** (0.22)	0.59*** (0.14)	0.38*** (0.14)	0.70*** (0.18)
Observations	287,181	65,025	71,606	74,138	76,412	536,009	123,316	133,757	137,800	141,136
ELA										
Katrina/Rita Share in Quartile 1	-1.37*** (0.35)	-1.29*** (0.41)	-0.28 (0.33)	-0.64 (0.41)	-0.16 (0.44)	-0.52*** (0.08)	-0.42*** (0.11)	-0.55*** (0.12)	-0.79*** (0.12)	-1.05*** (0.16)
Katrina/Rita Share in Quartile 2	0.29 (0.39)	1.28** (0.56)	0.14 (0.36)	-0.47 (0.30)	-0.99** (0.48)	0.15** (0.09)	1.13*** (0.17)	0.08 (0.13)	-0.24** (0.12)	-0.39 (0.15)
Katrina/Rita Share in Quartile 3	1.38*** (0.34)	2.12*** (0.67)	0.03 (0.39)	0.15 (0.30)	-0.25 (0.45)	0.15** (0.09)	1.11*** (0.20)	0.23* (0.13)	0.02 (0.12)	-0.21 (0.16)
Katrina/Rita Share in Quartile 4	1.73*** (0.41)	1.43* (0.73)	-0.15 (0.39)	0.48 (0.35)	0.80** (0.38)	0.52*** (0.11)	1.16*** (0.25)	0.50*** (0.15)	0.53*** (0.13)	0.66*** (0.17)
Observations	287,230	65,210	72,013	73,885	76,122	522,392	116,304	128,894	135,911	141,283

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 17 - Nonlinear Models of Evacuee Share and Evacuee Achievement on Incumbent Achievement - Louisiana
Evacuee Share by Test Administrator - Instrument with 2006 Evac Share for TA in Post-Katrina Years

	A. Elementary (No Lags)					B. Middle/High (Lags)				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Math										
Katrina/Rita Share in Quartile 1	-0.67** (0.28)	-0.71** (0.30)	-1.04*** (0.29)	-0.45 (0.29)	-0.65 (0.50)	-0.89*** (0.14)	-0.88*** (0.19)	-0.79*** (0.19)	-1.06*** (0.20)	-1.75*** (0.33)
Katrina/Rita Share in Quartile 2	-0.003 (0.34)	0.04 (0.62)	0.06 (0.32)	0.05 (0.34)	-1.02** (0.48)	0.05 (0.15)	1.12*** (0.26)	0.20 (0.21)	-0.11 (0.17)	-0.86** (0.34)
Katrina/Rita Share in Quartile 3	0.43 (0.37)	0.72 (0.51)	-0.25 (0.34)	-0.05 (0.41)	0.13 (0.57)	0.59*** (0.16)	1.85*** (0.29)	0.90*** (0.22)	0.47** (0.24)	-0.00 (0.28)
Katrina/Rita Share in Quartile 4	2.87*** (0.44)	1.10* (0.58)	1.38*** (0.43)	1.93*** (0.40)	2.20*** (0.58)	1.12*** (0.16)	2.18*** (0.42)	1.48*** (0.24)	0.79*** (0.25)	1.23*** (0.31)
Observations	287,181	65,025	71,606	74,138	76,412	526,563	119,343	131,439	135,994	139,787
ELA										
Katrina/Rita Share in Quartile 1	-1.37*** (0.35)	-1.29*** (0.41)	-0.28 (0.33)	-0.64 (0.41)	-0.16 (0.44)	-0.83*** (0.15)	-0.64*** (0.20)	-0.72*** (0.20)	-1.38*** (0.22)	-1.84*** (0.33)
Katrina/Rita Share in Quartile 2	0.29 (0.39)	1.28** (0.56)	0.14 (0.36)	-0.47 (0.30)	-0.99** (0.48)	0.27 (0.17)	1.93*** (0.33)	0.13 (0.23)	-0.26 (0.24)	-0.83*** (0.24)
Katrina/Rita Share in Quartile 3	1.38*** (0.34)	2.12*** (0.67)	0.03 (0.39)	0.15 (0.30)	-0.25 (0.45)	0.78*** (0.15)	2.89*** (0.35)	1.01*** (0.22)	0.33 (0.23)	0.10 (0.30)
Katrina/Rita Share in Quartile 4	1.73*** (0.41)	1.43* (0.73)	-0.15 (0.39)	0.48 (0.35)	0.80** (0.38)	1.30*** (0.19)	3.21*** (0.46)	1.26*** (0.29)	1.15*** (0.26)	1.43*** (0.28)
Observations	287,230	65,210	72,013	73,885	76,122	512,941	112,244	126,625	134,119	139,953

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 18 - Nonlinear Models of Evacuee Share in Grade and Evacuee Achievement on Incumbent Achievement - Louisiana

Alterincumbent Samples										
	A. Pooled Mid/High & Elem (No Lags, Full Sample)					B. Middle/High (No Lags, Lag Sample)				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Math										
Katrina/Rita Share in Quartile 1	0.63 (0.43)	0.04 (0.52)	-1.11*** (0.42)	-0.33 (0.49)	-0.57 (0.72)	-0.26 (0.46)	-0.89 (0.63)	-2.15*** (0.54)	-0.74 (0.53)	-1.25 (0.82)
Katrina/Rita Share in Quartile 2	-1.01* (0.57)	-1.18* (0.69)	-1.25** (0.58)	-1.34* (0.69)	-1.61 (1.07)	-0.93 (0.63)	-0.90 (0.81)	-1.17* (0.68)	-1.50* (0.79)	-1.22 (1.02)
Katrina/Rita Share in Quartile 3	1.09* (0.65)	1.07 (0.82)	1.06 (0.65)	1.72** (0.73)	0.95 (0.90)	0.40 (0.74)	-0.27 (0.99)	1.17 (0.81)	1.93** (0.85)	0.92 (1.16)
Katrina/Rita Share in Quartile 4	-0.65 (0.52)	0.51 (0.76)	1.57*** (0.56)	0.44 (0.59)	1.32* (0.70)	-0.17 (0.58)	1.79** (0.85)	2.00*** (0.66)	0.04 (0.60)	0.72 (0.92)
Observations	1,039,514	239,082	259,168	267,329	273,935	536,009	123,316	133,757	137,800	141,136
ELA										
Katrina/Rita Share in Quartile 1	-0.12 (0.41)	-0.91 (0.57)	-1.61*** (0.51)	-1.79*** (0.45)	-1.99*** (0.56)	-1.33** (0.53)	-1.85** (0.72)	-2.26*** (0.63)	-2.53*** (0.51)	-2.23*** (0.72)
Katrina/Rita Share in Quartile 2	0.10 (0.58)	0.78 (0.81)	-0.19 (0.58)	-1.11** (0.55)	-0.04 (0.76)	0.56 (0.69)	1.76* (1.01)	-0.02 (0.75)	-0.61 (0.72)	0.51 (0.96)
Katrina/Rita Share in Quartile 3	-0.41 (0.56)	-0.27 (0.92)	-0.11 (0.67)	-0.17 (0.58)	-0.57 (0.64)	-0.44 (0.74)	-0.66 (1.35)	-0.72 (0.84)	0.15 (0.78)	-1.02 (0.98)
Katrina/Rita Share in Quartile 4	-0.23 (0.44)	0.93 (0.91)	1.64*** (0.49)	2.28*** (0.49)	1.74*** (0.48)	0.20 (0.57)	0.73 (1.15)	2.24*** (0.68)	2.49*** (0.66)	1.66** (0.75)
Observations	1,023,498	231,026	253,886	264,940	273,646	522,392	116,304	128,894	135,911	141,283

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 19 - Nonlinear Models of Evacuee Share in Grade and Evacuee Achievement on Incumbent Achievement in Louisiana, Elementary (No Lags)

	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)
A. Math					
Katrina/Rita Share in Quartile 1	0.49 (0.56)	-1.45** (0.70)	-0.92* (0.56)	-1.80** (0.79)	0.10 (1.12)
Katrina/Rita Share in Quartile 2	-1.79** (0.73)	-0.66 (0.97)	-0.58 (0.92)	-1.34 (1.08)	-3.85** (1.57)
Katrina/Rita Share in Quartile 3	0.55 (0.89)	-0.08 (1.16)	-0.46 (1.00)	-0.34 (0.96)	-0.18 (1.33)
Katrina/Rita Share in Quartile 4	3.33*** (0.96)	3.24** (1.35)	2.83*** (0.97)	5.54*** (1.05)	5.17*** (1.20)
<i>Observations</i>	287,181	65,025	71,606	74,138	76,412
B. ELA					
Katrina/Rita Share in Quartile 1	0.34 (0.57)	-1.15 (0.93)	0.16 (0.62)	-1.04 (0.72)	-0.31 (0.78)
Katrina/Rita Share in Quartile 2	-1.33 (0.97)	-1.38 (1.44)	-1.09 (0.98)	-2.08** (1.01)	-2.05 (1.64)
Katrina/Rita Share in Quartile 3	-0.25 (0.68)	-0.32 (1.26)	-0.60 (0.83)	-0.29 (0.80)	-0.63 (0.94)
Katrina/Rita Share in Quartile 4	0.84 (0.64)	1.30 (1.64)	0.34 (0.75)	1.53* (0.86)	1.68** (0.76)
<i>Observations</i>	287,230	65,210	72,013	73,885	76,122

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year across the state. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, and grade-by-year fixed-effects. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 20- Tests of Models of Peer Effects Using Estimates from Table 7 - Bonferroni Adjusted

Model	A. Math				B. ELA			
	<i># of estimates significant in direction the model suggests</i>	<i># of estimates insignificant</i>	<i># of estimates significant in opposite direction</i>	<i>Consistent, Inconsistent, or neither?</i>	<i># of estimates significant in direction the model suggests</i>	<i># of estimates insignificant</i>	<i># of estimates significant in opposite direction</i>	<i>Consistent, Inconsistent, or neither?</i>
(1) Weak Monotonicity <i>Effect of evacuees from top quartile is statistically significantly greater than effect from bottom quartile.</i>	4	0	0	Consistent	3	1	0	Consistent
(2) Strong Monotonicity <i>Effect of evacuees from quartile j is statistically significantly greater than effect from quartile j - 1.</i>	0	12	0	Neither	1	11	0	Neither
(3) Invidious Comparison <i>Effect of evacuees from lower quartiles than the incumbent is positive and effect from evacuees in higher quartiles is negative.</i>	0	8	4	Inconsistent	0	10	2	Inconsistent
(4) Ability Grouping <i>Effect of evacuees from own quartile is statistically significantly greater than effect from higher quartiles.</i>	0	6	2	Inconsistent	0	6	2	Inconsistent

See text for the specific tests conducted for each model. Significance of tests is based on Bonferroni adjusted p-values to account for testing of multiple outcomes.

Appendix Table 21 - Nonlinear Models of Evacuee Share in Grade and Evacuee Achievement on Incumbent Achievement - Louisiana
Separate Time Trends for Schools with Above & Below Median Free/Reduced Price Lunch Rates in 2004-05

	A. Elementary (No Lags)					B. Middle/High (with Lags)				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Math										
Katrina/Rita Share in Quartile 1	0.39 (0.56)	-1.41** (0.70)	-1.23** (0.57)	-1.99** (0.80)	0.12 (1.14)	-0.66*** (0.25)	-0.68** (0.31)	-1.37*** (0.28)	-1.01*** (0.37)	-1.22** (0.54)
Katrina/Rita Share in Quartile 2	-1.82** (0.73)	-0.64 (0.97)	-0.71 (0.94)	-1.41 (1.09)	-3.85** (1.57)	-0.17 (0.31)	-0.56 (0.45)	-0.31 (0.39)	0.02 (0.38)	0.22 (0.61)
Katrina/Rita Share in Quartile 3	0.56 (0.89)	-0.09 (1.16)	-0.42 (1.02)	-0.32 (0.97)	-0.18 (1.33)	0.29 (0.31)	0.32 (0.47)	0.63 (0.39)	0.62 (0.39)	0.34 (0.49)
Katrina/Rita Share in Quartile 4	3.45*** (0.95)	3.20** (1.34)	3.16*** (1.01)	5.74*** (1.05)	5.15*** (1.20)	0.25 (0.30)	1.25** (0.54)	0.97** (0.40)	0.45 (0.37)	0.54 (0.54)
Observations	287,181	65,025	71,606	74,138	76,412	536,009	123,316	133,757	137,800	141,136
ELA										
Katrina/Rita Share in Quartile 1	0.18 (0.57)	-1.09 (0.93)	-0.10 (0.63)	-1.17 (0.73)	-0.28 (0.79)	-0.08 (0.29)	-0.34 (0.36)	-0.25 (0.39)	-0.78** (0.34)	-0.22 (0.43)
Katrina/Rita Share in Quartile 2	-1.39 (0.97)	-1.36 (1.44)	-1.19 (1.00)	-2.12** (1.02)	-2.04 (1.63)	0.06 (0.34)	0.80 (0.57)	-0.08 (0.47)	-0.30 (0.43)	-0.41 (0.41)
Katrina/Rita Share in Quartile 3	-0.28 (0.69)	-0.32 (1.25)	-0.67 (0.84)	-0.32 (0.81)	-0.63 (0.94)	-0.27 (0.35)	0.13 (0.58)	-0.11 (0.48)	-0.23 (0.41)	-0.12 (0.46)
Katrina/Rita Share in Quartile 4	1.03 (0.65)	1.23 (1.64)	0.66 (0.78)	1.66** (0.85)	1.65** (0.75)	0.27 (0.27)	1.44** (0.64)	0.84* (0.43)	1.55*** (0.35)	0.22 (0.39)
Observations	287,230	65,210	72,013	73,885	76,122	522,392	116,304	128,894	135,911	141,283

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 22 - Nonlinear Models of Evacuee Share in Grade and Evacuee Achievement on Incumbent Achievement - Louisiana
Separate Time Trends for Schools with Above & Below Median Evacuee Share in 2005-06

	A. Elementary (No Lags)					B. Middle/High (Lags)				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Math										
Katrina/Rita Share in Quartile 1	0.68 (0.57)	-1.35* (0.71)	-0.97* (0.57)	-1.73** (0.81)	0.06 (1.15)	-0.74*** (0.25)	-0.64** (0.31)	-1.47*** (0.29)	-1.29*** (0.39)	-1.68*** (0.54)
Katrina/Rita Share in Quartile 2	-1.71** (0.73)	-0.61 (0.97)	-0.60 (0.93)	-1.32 (1.08)	-3.87** (1.57)	-0.19 (0.31)	-0.52 (0.45)	-0.34 (0.40)	-0.10 (0.38)	0.02 (0.61)
Katrina/Rita Share in Quartile 3	0.61 (0.88)	-0.03 (1.15)	-0.48 (1.00)	-0.33 (0.96)	-0.19 (1.32)	0.37 (0.31)	0.36 (0.46)	0.75* (0.40)	0.74* (0.39)	0.48 (0.51)
Katrina/Rita Share in Quartile 4	3.43*** (0.96)	3.28** (1.35)	2.82*** (0.97)	5.58*** (1.05)	5.15*** (1.21)	0.43 (0.31)	1.27** (0.54)	1.26*** (0.41)	0.75* (0.39)	0.92* (0.56)
Observations	287,181	65,025	71,606	74,138	76,412	536,009	123,316	133,757	137,800	141,136
ELA										
Katrina/Rita Share in Quartile 1	0.50 (0.57)	-0.98 (0.91)	0.08 (0.63)	-0.95 (0.73)	-0.07 (0.79)	-0.04 (0.28)	-0.34 (0.36)	-0.31 (0.39)	-0.88** (0.35)	-0.45 (0.44)
Katrina/Rita Share in Quartile 2	-1.15 (0.97)	-1.24 (1.45)	-1.18 (0.99)	-1.97** (1.00)	-1.71 (1.62)	0.13 (0.34)	0.83 (0.57)	-0.05 (0.47)	-0.29 (0.43)	-0.46 (0.42)
Katrina/Rita Share in Quartile 3	-0.13 (0.68)	-0.22 (1.24)	-0.67 (0.84)	-0.23 (0.80)	-0.46 (0.92)	-0.14 (0.35)	0.26 (0.58)	0.08 (0.48)	-0.03 (0.42)	0.02 (0.48)
Katrina/Rita Share in Quartile 4	0.86 (0.63)	1.32 (1.62)	0.33 (0.75)	1.51* (0.85)	1.76** (0.76)	0.44 (0.28)	1.71*** (0.65)	1.16** (0.46)	1.87*** (0.38)	0.49 (0.40)
Observations	287,230	65,210	72,013	73,885	76,122	522,392	116,304	128,894	135,911	141,283

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 23 - Nonlinear Models of Evacuee Share in Grade and Evacuee Achievement on Incumbent Achievement - Louisiana
Including School-Specific Time Trends

	A. Elementary (No Lags)					B. Middle/High (with Lags)				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Math										
Katrina/Rita Share in Quartile 1	0.80 (0.71)	-1.94** (0.76)	-1.97*** (0.60)	-2.55*** (0.90)	-1.23 (1.38)	-0.61** (0.26)	-0.45 (0.32)	-1.11*** (0.32)	-1.08** (0.43)	-1.01* (0.57)
Katrina/Rita Share in Quartile 2	0.26 (0.95)	0.53 (1.13)	0.23 (1.12)	-0.41 (1.15)	-2.34 (1.97)	0.21 (0.33)	-0.18 (0.49)	-0.17 (0.42)	0.32 (0.41)	0.62 (0.63)
Katrina/Rita Share in Quartile 3	-0.03 (1.01)	0.70 (1.18)	-0.57 (1.14)	-0.81 (1.05)	-0.32 (1.66)	0.06 (0.31)	0.40 (0.53)	0.36 (0.38)	0.32 (0.42)	-0.17 (0.49)
Katrina/Rita Share in Quartile 4	4.90*** (1.07)	3.22** (1.39)	4.15*** (1.05)	6.56*** (1.19)	6.57*** (1.62)	0.08 (0.34)	0.73 (0.61)	0.50 (0.44)	0.38 (0.41)	0.14 (0.55)
Observations	287,181	65,025	71,606	74,138	76,412	536,009	123,316	133,757	137,800	141,136
ELA										
Katrina/Rita Share in Quartile 1	1.23* (0.69)	-0.84 (0.89)	-0.43 (0.76)	-1.74** (0.82)	-0.57 (0.98)	0.51* (0.31)	0.31 (0.40)	0.38 (0.46)	0.03 (0.37)	0.36 (0.44)
Katrina/Rita Share in Quartile 2	-1.48 (1.03)	-1.01 (1.51)	-1.92* (1.09)	-2.76** (1.10)	-2.27 (1.64)	0.43 (0.38)	1.01 (0.63)	0.22 (0.52)	0.06 (0.44)	-0.38 (0.45)
Katrina/Rita Share in Quartile 3	1.06 (0.77)	-0.13 (1.40)	-0.53 (0.98)	0.17 (0.93)	0.02 (1.00)	-0.08 (0.39)	0.34 (0.65)	0.02 (0.51)	-0.05 (0.43)	-0.14 (0.47)
Katrina/Rita Share in Quartile 4	0.82 (0.75)	1.57 (1.55)	1.69* (0.96)	3.07*** (0.88)	2.31*** (0.80)	0.09 (0.34)	1.07 (0.73)	0.36 (0.52)	0.92** (0.43)	-0.51 (0.42)
Observations	287,230	65,210	72,013	73,885	76,122	522,392	116,304	128,894	135,911	141,283

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 24 - Nonlinear Models of Evacuee Share in Grade and Evacuee Achievement on Incumbent Achievement - Louisiana
Including School-Specific Time Trends; Each Trend Set to Zero Post-Katrina

	A. Elementary (No Lags)					B. Middle/High (with Lags)				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Math										
Katrina/Rita Share in Quartile 1	0.41 (0.67)	-1.13 (0.78)	-0.52 (0.65)	-1.45* (0.87)	0.78 (1.11)	-0.60** (0.25)	-0.23 (0.30)	-0.80*** (0.30)	-0.52 (0.41)	-0.56 (0.54)
Katrina/Rita Share in Quartile 2	-2.60*** (0.89)	-1.49 (1.17)	-0.96 (0.99)	-1.56 (1.19)	-4.10*** (1.52)	0.11 (0.32)	-0.11 (0.49)	0.03 (0.42)	0.56 (0.39)	0.80 (0.60)
Katrina/Rita Share in Quartile 3	0.97 (1.10)	0.15 (1.46)	0.23 (1.15)	0.15 (1.05)	-0.04 (1.36)	0.34 (0.31)	0.37 (0.50)	0.42 (0.38)	0.56 (0.40)	0.36 (0.50)
Katrina/Rita Share in Quartile 4	2.44* (1.26)	2.83* (1.65)	1.80 (1.10)	4.67*** (1.11)	4.00*** (1.21)	0.40 (0.34)	1.01* (0.58)	0.38 (0.44)	0.25 (0.39)	0.21 (0.60)
Observations	287,181	65,025	71,606	74,138	76,412	536,009	123,316	133,757	137,800	141,136
Katrina/Rita Share in Quartile 1	0.07 (0.67)	-0.64 (1.12)	0.60 (0.68)	-0.55 (0.77)	-0.05 (0.79)	0.39 (0.31)	0.47 (0.41)	0.63 (0.45)	0.07 (0.38)	0.83* (0.42)
Katrina/Rita Share in Quartile 2	-1.32 (1.08)	-0.69 (1.64)	-0.72 (1.14)	-2.18** (1.06)	-1.86 (1.67)	0.29 (0.37)	1.06* (0.60)	0.31 (0.50)	0.18 (0.47)	-0.43 (0.43)
Katrina/Rita Share in Quartile 3	-0.50 (0.84)	-1.18 (1.45)	-0.50 (0.95)	-0.33 (0.86)	-0.57 (1.07)	0.04 (0.36)	0.38 (0.63)	0.13 (0.48)	-0.06 (0.41)	0.04 (0.41)
Katrina/Rita Share in Quartile 4	0.64 (0.71)	0.92 (1.99)	-0.39 (0.76)	0.88 (0.93)	1.14 (0.78)	0.31 (0.33)	1.39* (0.73)	0.51 (0.49)	1.01** (0.40)	-0.82** (0.36)
Observations	287,230	65,210	72,013	73,885	76,122	522,392	116,304	128,894	135,911	141,283

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with the number of years between exams. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 25 - School Level Placebo Test Regressions of School Performance Score on Evacuee Shares in Louisiana - 1999-00 Through 2004-05, No Controls

	School Performance Scores in Year t Regressed on Evacuee Shares in Year t + 2	School Performance Scores in Year t Regressed on Evacuee Shares in Year t + 3
	(1)	(2)
Katrina/Rita Share	-5.7 (4.1)	-3.36 (2.9)
<i>Observations</i>	5,738	4,795
Katrina/Rita Share in Quartile 1	-56.0** (21.7)	-29.3 (23.8)
Katrina/Rita Share in Quartile 2	-23.2 (35.1)	-7.0 (27.9)
Katrina/Rita Share in Quartile 3	32.5 (30.6)	-15.1 (30.7)
Katrina/Rita Share in Quartile 4	9.8 (22.9)	0.7 (20.0)
<i>Observations</i>	5,738	4,795
Mean School Performance Score		83.1
Standard Deviation of SPS		20.9

Robust standard errors clustered by school are provided in parentheses and clustered by school. All regressions also include school fixed-effects and year dummies. School performance score (SPS) is a metric used to determine accountability ratings that includes test scores, attendance rates, dropouts, and graduation rates. See text for more details. In column (1), evacuee shares in 2005-06 and 2006-07 are matched to the same schools in 2003-04 and 2004-05, respectively so that all SPS scores in the regression are pre-Katrina. For 1999-00 through 2002-03 evacuee shares are set to zero. For column (2) the analysis is conducted one year earlier. Quartiles for evacuees in LA are within grade across the state data and determined from their most recent pre-Katrina score. Since SPS is an aggregate measure that includes both math and reading scores, evacuee shares are averaged across math and reading quartiles. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 26 - School Level Pre-Trend Regressions of School Performance Score on Evacuee Shares in Louisiana - 1999-00 Through 2004-05

	A. Overall			B. By Evacuee Quartile	
	Without Controls	With Controls		Without Controls	With Controls
	(1)	(2)		(1)	(2)
Katrina/Rita Share*2000-01	4.6 (4.8)	1.8 (5.0)	Katrina/Rita Share in Q1*2000-01	35.6 (28.9)	13.1 (26.3)
Katrina/Rita Share*2000-02	-0.5 (5.9)	-2.3 (4.9)	Katrina/Rita Share in Q1*2000-02	4.2 (25.7)	-9.4 (22.1)
Katrina/Rita Share*2002-03	-1.1 (6.3)	-3.4 (4.4)	Katrina/Rita Share in Q1*2002-03	0.6 (24.3)	-13.0 (22.1)
Katrina/Rita Share*2003-04	-5.6 (6.9)	-4.7 (5.7)	Katrina/Rita Share in Q1*2003-04	-31.8 (26.6)	-28.0 (24.5)
Katrina/Rita Share*2004-05	-10.0 (6.9)	-7.1 (6.7)	Katrina/Rita Share in Q1*2004-05	-65.0** (30.7)	-76.9*** (26.7)
Observations	4,978	4,978	Katrina/Rita Share in Q2*2000-01	24.1 (45.9)	17.9 (39.3)
			Katrina/Rita Share in Q2*2000-02	40.5 (44.0)	59.8 (39.1)
			Katrina/Rita Share in Q2*2002-03	17.2 (43.8)	42.8 (39.9)
			Katrina/Rita Share in Q2*2003-04	-9.0 (51.7)	-18.0 (47.4)
			Katrina/Rita Share in Q2*2004-05	4.0 (61.5)	16.9 (55.4)
			Katrina/Rita Share in Q3*2000-01	-7.6 (35.5)	7.8 (36.9)
			Katrina/Rita Share in Q3*2000-02	-10.0 (46.4)	-13.3 (37.5)
			Katrina/Rita Share in Q3*2002-03	-0.1 (53.6)	-25.1 (40.3)
			Katrina/Rita Share in Q3*2003-04	18.6 (59.5)	25.0 (47.2)
			Katrina/Rita Share in Q3*2004-05	-11.5 (63.4)	2.0 (53.5)
			Katrina/Rita Share in Q4*2000-01	-26.5 (25.1)	-27.8 (24.5)
			Katrina/Rita Share in Q4*2000-02	-28.0 (32.0)	-34.5 (24.9)
			Katrina/Rita Share in Q4*2002-03	-17.5 (37.3)	-9.8 (27.3)
			Katrina/Rita Share in Q4*2003-04	-2.4 (44.7)	-2.9 (33.2)
			Katrina/Rita Share in Q4*2004-05	27.4 (51.9)	22.2 (40.3)
			Observations	4,978	4,978

Robust standard errors clustered by school are provided in parentheses and clustered by school. All regressions also include school fixed-effects and year dummies. Control regressions also include % of tested students who are African-American, Hispanic, Asian, free/reduced-price lunch, and the distribution of tested students across grades in each year. School performance score (SPS) is a metric used to determine accountability ratings that includes test scores, attendance rates, dropouts, and graduation rates. See text for more details. In column (1), evacuee shares in 2005-06 and 2006-07 are matched to the same schools in 2003-04 and 2004-05, respectively so that all SPS scores in the regression are pre-Katrina. For 1999-00 through 2002-03 evacuee shares are set to zero. For column (2) the analysis is conducted one year earlier. Quartiles for evacuees in LA are within grade across the state data and determined from their most recent pre-Katrina score. Since SPS is an aggregate measure that includes both math and reading scores, evacuee shares are averaged across math and reading quartiles. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 27 - Nonlinear Models of Evacuee Share in Grade and Evacuee Achievement on Incumbent Achievement - Louisiana
Excluding 2004-05 (Interrupted Panel)

	A. Elementary (No Lags)					B. Middle/High (Lags)				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Math										
Katrina/Rita Share in Quartile 1	0.38 (0.60)	-1.58** (0.72)	-0.69 (0.59)	-1.46* (0.80)	0.42 (1.09)	-0.86*** (0.25)	-0.57* (0.31)	-1.32*** (0.29)	-1.08*** (0.39)	-1.41*** (0.54)
Katrina/Rita Share in Quartile 2	-2.04** (0.80)	-0.55 (1.10)	-0.65 (0.94)	-1.34 (1.09)	-3.73** (1.53)	-0.26 (0.32)	-0.48 (0.46)	-0.31 (0.40)	0.05 (0.38)	0.14 (0.61)
Katrina/Rita Share in Quartile 3	0.54 (0.96)	-0.09 (1.26)	-0.24 (1.03)	-0.26 (0.96)	-0.10 (1.30)	0.33 (0.31)	0.30 (0.47)	0.56 (0.39)	0.70* (0.39)	0.48 (0.50)
Katrina/Rita Share in Quartile 4	3.12*** (1.07)	2.89** (1.43)	2.43** (0.98)	5.08*** (1.05)	4.49*** (1.16)	0.51 (0.31)	1.20** (0.57)	1.00** (0.41)	0.60 (0.38)	0.71 (0.55)
Observations	243,105	54,889	60,610	62,838	64,768	471,601	109,539	117,454	120,740	123,868
ELA										
Katrina/Rita Share in Quartile 1	0.13 (0.61)	-1.05 (1.00)	0.39 (0.63)	-0.82 (0.74)	-0.28 (0.77)	-0.10 (0.29)	-0.15 (0.37)	-0.09 (0.41)	-0.70* (0.36)	-0.10 (0.43)
Katrina/Rita Share in Quartile 2	-1.13 (0.99)	-1.02 (1.47)	-0.94 (1.02)	-2.06** (1.02)	-1.88 (1.55)	0.07 (0.35)	0.87 (0.57)	-0.01 (0.48)	-0.20 (0.45)	-0.50 (0.42)
Katrina/Rita Share in Quartile 3	-0.44 (0.78)	-0.53 (1.33)	-0.54 (0.85)	-0.16 (0.82)	-0.66 (0.95)	-0.17 (0.35)	0.23 (0.59)	0.03 (0.48)	-0.07 (0.41)	0.06 (0.45)
Katrina/Rita Share in Quartile 4	0.80 (0.68)	1.03 (1.77)	0.05 (0.74)	1.23 (0.86)	1.55** (0.76)	0.32 (0.28)	1.50** (0.67)	0.85* (0.46)	1.53*** (0.36)	0.07 (0.38)
Observations	243,149	55,035	60,992	62,555	64,567	460,245	103,818	113,413	119,098	123,916

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the

Appendix Table 28 - Nonlinear Models of Evacuee Share in Grade and Evacuee Achievement on Incumbent Achievement - Louisiana
Limited To Schools With < 10% Evacuees in 2006

	A. Elementary (No Lags)					B. Middle/High (Lags)				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Math										
Katrina/Rita Share in Quartile 1	0.76 (0.65)	-1.45* (0.84)	-0.70 (0.60)	-1.62* (0.92)	0.37 (1.30)	-0.83*** (0.31)	-0.56 (0.40)	-1.86*** (0.37)	-1.41*** (0.46)	-1.43** (0.64)
Katrina/Rita Share in Quartile 2	-1.82** (0.88)	-1.01 (1.15)	-0.78 (0.99)	-1.03 (1.27)	-4.46*** (1.67)	-0.17 (0.35)	-0.55 (0.51)	-0.59 (0.46)	0.08 (0.43)	0.11 (0.68)
Katrina/Rita Share in Quartile 3	-0.15 (1.11)	-1.45 (1.44)	-1.89 (1.18)	-1.05 (1.18)	0.16 (1.75)	0.35 (0.33)	0.23 (0.58)	0.67 (0.42)	0.88** (0.43)	0.51 (0.54)
Katrina/Rita Share in Quartile 4	4.07*** (1.27)	2.79* (1.68)	2.57** (1.24)	6.32*** (1.31)	7.18*** (1.40)	0.53 (0.36)	1.19* (0.61)	1.32*** (0.45)	0.64 (0.46)	0.87 (0.67)
Observations	273,171	61,741	68,248	70,532	72,650	512,186	117,548	127,961	131,910	134,767
ELA										
Katrina/Rita Share in Quartile 1	0.78 (0.62)	-1.21 (1.19)	0.14 (0.66)	-1.19 (0.84)	-0.36 (0.96)	-0.13 (0.31)	-0.48 (0.41)	-0.48 (0.41)	-1.20*** (0.41)	-0.29 (0.52)
Katrina/Rita Share in Quartile 2	-1.92* (1.07)	-2.35 (1.53)	-1.53 (1.10)	-2.38** (1.09)	-2.94 (1.86)	-0.04 (0.38)	1.00* (0.61)	-0.38 (0.47)	-0.47 (0.48)	-0.87* (0.44)
Katrina/Rita Share in Quartile 3	-0.84 (0.87)	-0.92 (1.67)	-1.22 (1.05)	-0.68 (1.02)	-1.18 (1.18)	-0.32 (0.37)	-0.22 (0.65)	-0.15 (0.49)	0.02 (0.43)	-0.20 (0.49)
Katrina/Rita Share in Quartile 4	1.48 (0.98)	0.91 (2.16)	0.14 (1.13)	3.28*** (0.93)	3.17** (1.26)	0.62** (0.31)	1.87** (0.74)	1.22*** (0.44)	1.78*** (0.39)	0.69 (0.47)
Observations	273,217	62,084	68,808	70,214	72,111	499,208	111,303	123,465	129,781	134,659

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with the number of years between exams.

Appendix Table 29 - Nonlinear Models of Evacuee Share in Grade and Evacuee Achievement on Incumbent Achievement - Louisiana
Unstandardized Scale Scores

	A. Elementary (No Lags)					B. Middle/High (Lags)				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Math										
Katrina/Rita Share in Quartile 1	-55 (35)	-110** (44)	-78** (35)	-126** (50)	-10 (73)	-66*** (16)	-37** (18)	-93*** (19)	-90*** (24)	-108*** (36)
Katrina/Rita Share in Quartile 2	-131*** (50)	-39 (61)	-29 (63)	-99 (74)	-264** (108)	-24 (19)	-23 (25)	-21 (24)	-3 (24)	-14 (39)
Katrina/Rita Share in Quartile 3	42 (56)	-12 (70)	-30 (62)	-10 (61)	-6 (90)	40** (20)	18 (27)	63*** (24)	52** (25)	59* (33)
Katrina/Rita Share in Quartile 4	302*** (60)	216*** (83)	200*** (60)	359*** (67)	363*** (80)	60*** (23)	94*** (32)	63** (24)	53** (26)	74** (36)
Observations	287,181	65,059	71,675	74,020	76,427	536,009	122,764	133,719	137,944	141,582
ELA										
Katrina/Rita Share in Quartile 1	3 (30)	-68 (50)	5 (34)	-64* (39)	-21 (42)	-25 (17)	-21 (19)	-18 (22)	-58*** (19)	-53* (27)
Katrina/Rita Share in Quartile 2	-71 (51)	-70 (78)	-69 (53)	-89 (55)	-102 (88)	-16 (22)	5 (31)	-20 (27)	-31 (25)	-21 (25)
Katrina/Rita Share in Quartile 3	-21 (37)	-24 (69)	-26 (45)	-29 (44)	-40 (50)	3 (23)	32 (30)	-11 (30)	2 (25)	9 (31)
Katrina/Rita Share in Quartile 4	65* (34)	75 (89)	30 (40)	86* (47)	94** (41)	66*** (18)	104*** (35)	76*** (26)	107*** (20)	49* (25)
Observations	287,201	65,242	72,024	73,842	76,122	522,361	116,469	128,331	136,073	141,519

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 30 - Nonlinear Models of Evacuee Share in Grade and Evacuee Achievement on Incumbent Achievement - Louisiana
Instrument for Evacuee Shares with Share in Student's Most Recent Pre-2006 School

	A. Elementary (No Lags)					B. Middle/High (with Lags)				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Math										
Katrina/Rita Share in Quartile 1	-0.83 (1.12)	-2.03 (1.43)	-2.10 (1.54)	-3.84*** (1.33)	-0.96 (1.78)	-0.15 (1.06)	-1.38 (1.25)	-3.19*** (1.09)	-2.22* (1.28)	-1.66 (2.52)
Katrina/Rita Share in Quartile 2	-3.71** (1.70)	-3.97 (2.43)	-2.06 (2.05)	-5.55** (2.40)	-8.36** (3.61)	-4.97*** (1.25)	-4.07** (1.69)	-2.94** (1.32)	-5.74*** (1.52)	-7.79*** (2.22)
Katrina/Rita Share in Quartile 3	2.37* (1.42)	4.37* (2.43)	-0.93 (2.29)	0.98 (2.11)	1.87 (2.13)	2.71* (1.44)	3.79* (2.17)	4.53*** (1.43)	1.14 (1.80)	3.41 (2.40)
Katrina/Rita Share in Quartile 4	4.69** (2.06)	4.93* (2.53)	4.59* (2.69)	10.46*** (2.56)	7.59*** (2.52)	3.64*** (1.13)	4.61** (1.93)	3.89*** (1.46)	5.55*** (1.39)	5.64*** (1.81)
Observations	287,181	65,016	71,628	74,111	76,426	536,009	123,282	133,685	137,881	141,161
ELA										
Katrina/Rita Share in Quartile 1	-1.17 (0.96)	-2.79* (1.49)	-1.32 (1.75)	-2.22* (1.27)	-2.02 (1.46)	-2.45*** (0.89)	-3.37** (1.38)	-1.71 (1.07)	-2.05** (1.01)	-3.48*** (1.24)
Katrina/Rita Share in Quartile 2	-0.92 (1.77)	-0.80 (2.93)	-1.56 (1.94)	-2.11 (2.19)	-3.47 (2.32)	-0.83 (1.17)	1.25 (1.78)	-1.62 (1.49)	-2.95** (1.36)	-0.86 (1.67)
Katrina/Rita Share in Quartile 3	-0.76 (1.27)	-0.80 (2.21)	-0.74 (2.31)	-1.57 (1.75)	-0.98 (2.05)	-1.78 (1.27)	-2.44 (1.94)	-0.42 (1.52)	-1.47 (1.59)	-1.86 (1.79)
Katrina/Rita Share in Quartile 4	1.93 (1.29)	4.50** (2.20)	0.61 (2.60)	4.15*** (1.60)	3.40** (1.54)	5.11*** (1.12)	6.54*** (1.94)	5.35*** (1.97)	5.76*** (1.39)	5.11*** (1.50)
Observations	287,201	65,031	71,628	74,115	76,427	522,361	115,257	129,879	136,476	140,749

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with the number of years between exams. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 31 - Nonlinear Models of Evacuee Share in School and Evacuee Achievement on Incumbent Achievement - Louisiana, Including Student Fixed-Effects - Pooled Sample (No Lags)

	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)
Math					
Katrina/Rita Share in Quartile 1	-0.76*** (0.19)	-0.98*** (0.26)	-1.41*** (0.25)	-1.59*** (0.30)	-1.33*** (0.41)
Katrina/Rita Share in Quartile 2	-0.29 (0.24)	-0.51 (0.39)	-0.51* (0.31)	-0.17 (0.32)	-0.44 (0.49)
Katrina/Rita Share in Quartile 3	0.67*** (0.25)	1.24*** (0.43)	0.89*** (0.32)	0.72*** (0.33)	1.01** (0.43)
Katrina/Rita Share in Quartile 4	0.89*** (0.32)	1.27*** (0.44)	1.16*** (0.36)	1.52*** (0.37)	1.79*** (0.51)
<i>Observations</i>	1,039,513	239,081	259,168	267,329	273,935
ELA					
Katrina/Rita Share in Quartile 1	-0.18 (0.19)	-0.33 (0.30)	-0.85*** (0.29)	-1.00*** (0.27)	-0.74** (0.30)
Katrina/Rita Share in Quartile 2	-0.08 (0.25)	0.57 (0.44)	-0.17 (0.35)	-0.60** (0.30)	-0.77** (0.36)
Katrina/Rita Share in Quartile 3	-0.29 (0.25)	-0.01 (0.51)	0.16 (0.35)	-0.27 (0.32)	0.02 (0.36)
Katrina/Rita Share in Quartile 4	0.84*** (0.22)	2.03*** (0.53)	1.72*** (0.37)	2.20*** (0.28)	1.16*** (0.30)
<i>Observations</i>	1,023,497	231,025	253,886	264,940	273,646

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, and grade-by-year fixed-effects. Includes grade 4 from 1999-2000 through 2004-2005, grade 5 in 2005-2006, grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 32 - Nonlinear Models of Evacuee Share in Grade and Evacuee Achievement on Incumbent Achievement - Louisiana, School-Grade FE's, Middle/High

	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(6)	(7)	(8)	(9)	(10)
A. Math					
Katrina/Rita Share in Quartile 1	-0.51* (0.26)	-0.74* (0.40)	-1.20*** (0.33)	-0.78** (0.37)	-1.48*** (0.54)
Katrina/Rita Share in Quartile 2	-0.27 (0.32)	-0.58 (0.56)	-0.34 (0.45)	-0.09 (0.42)	0.12 (0.64)
Katrina/Rita Share in Quartile 3	-0.25 (0.34)	-0.28 (0.56)	-0.03 (0.45)	0.18 (0.43)	-0.01 (0.68)
Katrina/Rita Share in Quartile 4	0.07 (0.33)	1.08 (0.66)	0.73* (0.43)	0.68 (0.42)	0.54 (0.61)
Observations	536,009	123,316	133,757	137,800	141,136
B. ELA					
Katrina/Rita Share in Quartile 1	0.19 (0.28)	-0.64 (0.42)	0.17 (0.36)	-0.56 (0.39)	-0.32 (0.47)
Katrina/Rita Share in Quartile 2	0.02 (0.32)	0.70 (0.65)	-0.16 (0.44)	-0.27 (0.44)	-0.26 (0.51)
Katrina/Rita Share in Quartile 3	-0.21 (0.33)	0.00 (0.61)	-0.28 (0.44)	0.17 (0.46)	-0.01 (0.60)
Katrina/Rita Share in Quartile 4	0.09 (0.29)	0.79 (0.75)	0.64 (0.48)	1.30*** (0.44)	0.58 (0.49)
Observations	522,392	116,304	128,894	135,911	141,283

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with the number of years between exams. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 33 - Nonlinear Models of Evacuee Share in Grade and Evacuee Achievement on Incumbent Achievement - Louisiana; Middle/High, Lags Not Interacted with Number of Years Between Scores

	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)
Math					
Katrina/Rita Share in Quartile 1	-0.81*** (0.25)	-0.69** (0.31)	-1.55*** (0.29)	-1.29*** (0.38)	-1.68*** (0.56)
Katrina/Rita Share in Quartile 2	-0.24 (0.31)	-0.58 (0.45)	-0.41 (0.40)	-0.15 (0.39)	-0.04 (0.62)
Katrina/Rita Share in Quartile 3	0.26 (0.30)	0.30 (0.46)	0.68* (0.40)	0.75* (0.40)	0.45 (0.51)
Katrina/Rita Share in Quartile 4	0.39 (0.30)	1.28** (0.54)	1.23*** (0.41)	0.74* (0.39)	1.01* (0.57)
<i>Observations</i>	536,009	123,316	133,757	137,800	141,136
ELA					
Katrina/Rita Share in Quartile 1	-0.17 (0.28)	-0.47 (0.36)	-0.46 (0.39)	-1.00*** (0.35)	-0.63 (0.46)
Katrina/Rita Share in Quartile 2	0.08 (0.35)	0.75 (0.57)	-0.19 (0.48)	-0.45 (0.44)	-0.43 (0.44)
Katrina/Rita Share in Quartile 3	-0.33 (0.35)	0.23 (0.58)	-0.05 (0.48)	-0.14 (0.42)	-0.01 (0.49)
Katrina/Rita Share in Quartile 4	0.36 (0.27)	1.68*** (0.65)	1.13** (0.45)	1.83*** (0.36)	0.50 (0.41)
<i>Observations</i>	522,392	116,304	128,894	135,911	141,283

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 34 - Nonlinear Models of Evacuee Number in Grade and Evacuee Achievement on Incumbent Achievement While Controlling for Quartic in Total Number in Grade Tested - Louisiana

	A. Elementary (No Lags)					B. Middle/High (Lags)				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Math										
Katrina/Rita Share in Quartile 1	0.01818** (0.00924)	-0.00117 (0.01145)	-0.00618 (0.00965)	-0.01609 (0.01164)	0.00139 (0.01762)	-0.00466** (0.00202)	-0.00399 (0.00248)	-0.01080*** (0.00247)	-0.00831*** (0.00249)	-0.00957** (0.00374)
Katrina/Rita Share in Quartile 2	-0.00742 (0.00905)	-0.01273 (0.01299)	-0.00284 (0.01061)	-0.00932 (0.01119)	-0.01969 (0.01901)	0.00030 (0.00277)	-0.00135 (0.00356)	-0.00197 (0.00305)	-0.00300 (0.00341)	-0.00109 (0.00440)
Katrina/Rita Share in Quartile 3	0.00133 (0.00998)	-0.00031 (0.01674)	0.00163 (0.01091)	0.00331 (0.01173)	0.00095 (0.01404)	0.00420* (0.00224)	0.00344 (0.00355)	0.00890*** (0.00336)	0.00715*** (0.00267)	0.00371 (0.00311)
Katrina/Rita Share in Quartile 4	0.01299 (0.00874)	0.03077** (0.01317)	0.02024** (0.00890)	0.03304*** (0.01006)	0.02711** (0.01272)	0.00345 (0.00220)	0.01033*** (0.00341)	0.00857*** (0.00314)	0.00635** (0.00252)	0.00698** (0.00280)
Observations	287,181	65,025	71,606	74,138	76,412	536,009	123,316	133,757	137,800	141,136
ELA										
Katrina/Rita Share in Quartile 1	0.01300 (0.00860)	-0.01660 (0.01515)	-0.00076 (0.00914)	-0.01551 (0.01216)	-0.00916 (0.01061)	0.00281 (0.00193)	-0.00069 (0.00296)	0.00102 (0.00233)	-0.00432* (0.00238)	-0.00421 (0.00268)
Katrina/Rita Share in Quartile 2	-0.00159 (0.01223)	0.00950 (0.01995)	0.00080 (0.01464)	-0.00944 (0.01424)	-0.00732 (0.01838)	0.00135 (0.00210)	0.00451 (0.00404)	-0.00068 (0.00268)	-0.00102 (0.00299)	-0.00132 (0.00270)
Katrina/Rita Share in Quartile 3	-0.00825 (0.00741)	-0.01018 (0.01286)	-0.00369 (0.00903)	0.00413 (0.00877)	-0.00229 (0.01009)	-0.00178 (0.00226)	0.00086 (0.00431)	-0.00200 (0.00293)	-0.00146 (0.00336)	0.00060 (0.00228)
Katrina/Rita Share in Quartile 4	-0.00176 (0.00523)	0.01935 (0.01461)	0.00361 (0.00689)	0.00963 (0.00619)	0.00735 (0.00647)	-0.00084 (0.00164)	0.00277 (0.00357)	0.00634*** (0.00224)	0.00877*** (0.00209)	0.00340 (0.00217)
Observations	287,230	65,210	72,013	73,885	76,122	522,392	116,304	128,894	135,911	141,283

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with the number of years between exams. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 35 - Nonlinear Models of Evacuee Share in Grade and Evacuee Achievement on Incumbent Achievement - Louisiana
Controlling for % Of Evacuees in Grade Who Are Black & % of Evacuees in Grade Who Qualify for Free/Reduced-Price Lunch

	A. Elementary (No Lags)					B. Middle/High (with Lags)				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Math										
Katrina/Rita Share in Quartile 1	1.13*	-0.94	-0.10	-0.88	1.39	-0.67***	-0.41	-1.18***	-0.90**	-1.66***
	(0.64)	(0.77)	(0.62)	(0.87)	(1.26)	(0.25)	(0.31)	(0.30)	(0.38)	(0.58)
Katrina/Rita Share in Quartile 2	-1.55**	-0.46	-0.22	-1.01	-3.38**	-0.18	-0.41	-0.23	0.05	0.03
	(0.75)	(0.99)	(0.95)	(1.08)	(1.52)	(0.32)	(0.44)	(0.40)	(0.39)	(0.63)
Katrina/Rita Share in Quartile 3	0.44	-0.25	-0.62	-0.47	-0.35	0.36	0.40	0.74*	0.76*	0.49
	(0.89)	(1.14)	(0.96)	(0.96)	(1.37)	(0.31)	(0.47)	(0.40)	(0.40)	(0.51)
Katrina/Rita Share in Quartile 4	2.55***	2.38*	1.62*	4.53***	4.03***	0.37	1.19**	1.03**	0.56	0.93*
	(0.95)	(1.34)	(0.96)	(1.09)	(1.21)	(0.31)	(0.54)	(0.40)	(0.40)	(0.56)
Observations	287,181	65,025	71,606	74,138	76,412	536,009	123,316	133,757	137,800	141,136
ELA										
Katrina/Rita Share in Quartile 1	0.51	-0.70	0.68	-0.75	0.28	-0.12	-0.54	-0.26	-0.62*	-0.31
	(0.61)	(0.95)	(0.68)	(0.78)	(0.81)	(0.29)	(0.36)	(0.39)	(0.35)	(0.46)
Katrina/Rita Share in Quartile 2	-1.29	-1.03	-0.77	-2.09**	-1.80	0.03	0.61	-0.09	-0.22	-0.45
	(0.96)	(1.47)	(0.97)	(1.01)	(1.52)	(0.34)	(0.57)	(0.47)	(0.43)	(0.43)
Katrina/Rita Share in Quartile 3	-0.38	-0.72	-0.67	-0.49	-0.71	-0.24	0.09	0.01	-0.07	-0.02
	(0.69)	(1.23)	(0.81)	(0.80)	(0.93)	(0.35)	(0.58)	(0.48)	(0.42)	(0.47)
Katrina/Rita Share in Quartile 4	0.56	0.88	-0.06	1.11	0.98	0.32	1.51**	0.99**	1.59***	0.36
	(0.63)	(1.61)	(0.74)	(0.84)	(0.66)	(0.27)	(0.65)	(0.45)	(0.36)	(0.40)
Observations	287,230	65,210	72,013	73,885	76,122	522,392	116,304	128,894	135,911	141,283

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with the number of years between exams. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 36 - Nonlinear Models of Evacuee Share in Grade and Evacuee Achievement on Incumbent Achievement - Louisiana, Quintiles

	A. Elementary (No Lags)					B. Middle/High (Lags)				
	Bottom Quintile	2nd Quintile	3rd Quintile	4th Quintile	Top Quintile	Bottom Quintile	2nd Quintile	3rd Quintile	4th Quintile	Top Quintile
Pre-Katrina LEAP Quintile	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Math										
Katrina/Rita Share in Quintile 1	-2.13** (0.85)	-1.22* (0.63)	-0.78 (0.76)	-1.92** (0.91)	-0.80 (1.28)	-0.75** (0.37)	-1.43*** (0.33)	-1.93*** (0.36)	-0.49 (0.47)	-1.91*** (0.64)
Katrina/Rita Share in Quintile 2	1.51 (1.26)	-0.05 (0.97)	0.28 (1.10)	0.19 (1.51)	-2.79 (1.83)	-0.48 (0.57)	-1.06** (0.43)	-0.60 (0.41)	-1.03** (0.49)	-0.08 (0.69)
Katrina/Rita Share in Quintile 3	-1.89 (1.39)	-0.59 (1.13)	-0.98 (1.15)	-0.53 (1.26)	-0.03 (1.82)	-0.52 (0.53)	0.72 (0.57)	0.20 (0.48)	0.43 (0.45)	0.30 (0.67)
Katrina/Rita Share in Quintile 4	-0.79 (1.62)	-0.66 (1.09)	-0.20 (1.40)	-0.09 (1.15)	-1.78 (1.78)	1.25* (0.64)	0.87** (0.44)	1.07** (0.52)	0.61 (0.46)	0.44 (0.58)
Katrina/Rita Share in Quintile 5	4.56*** (1.55)	2.88** (1.18)	3.35*** (1.23)	5.73*** (1.10)	6.02*** (1.63)	1.23* (0.74)	1.36** (0.55)	1.24** (0.50)	0.47 (0.46)	1.34* (0.78)
Observations	51,186	56,469	58,398	59,656	61,472	97,055	106,010	108,652	110,924	113,368
ELA										
Katrina/Rita Share in Quintile 1	-0.82 (1.12)	0.49 (0.78)	-1.83*** (0.67)	-1.23 (0.90)	-1.00 (1.03)	-0.54 (0.41)	-0.54 (0.41)	-0.47 (0.43)	-1.38*** (0.39)	-0.16 (0.55)
Katrina/Rita Share in Quintile 2	-1.07 (1.52)	0.51 (1.17)	-0.59 (1.07)	-0.89 (1.29)	0.50 (1.29)	0.56 (0.68)	-0.08 (0.58)	-1.13** (0.56)	-0.11 (0.49)	-0.98** (0.48)
Katrina/Rita Share in Quintile 3	-4.81*** (1.73)	-1.48 (1.21)	-0.91 (1.21)	-2.36* (1.30)	-3.23 (2.35)	0.12 (0.62)	0.36 (0.48)	-0.14 (0.51)	-0.45 (0.53)	0.56 (0.67)
Katrina/Rita Share in Quintile 4	2.82* (1.65)	0.49 (1.14)	-0.01 (1.01)	2.22** (1.11)	0.81 (1.49)	0.70 (0.93)	0.18 (0.58)	0.11 (0.53)	-0.33 (0.56)	-0.51 (0.58)
Katrina/Rita Share in Quintile 5	0.34 (1.93)	-0.04 (1.10)	1.10 (1.02)	0.83 (0.82)	1.08 (0.87)	2.09** (0.91)	1.56*** (0.57)	1.76*** (0.50)	2.10*** (0.50)	0.71 (0.54)
Observations	51,295	56,874	58,492	59,380	61,189	91,930	100,486	106,503	109,992	113,481

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quintiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with the number of years between exams. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 37 - Nonlinear Models of Evacuee Share in Grade and Evacuee Achievement on Incumbent Achievement - Louisiana
School-Year FE's - Limited to 2005-06 and Later, Middle/High

	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(6)	(7)	(8)	(9)	(10)
A. Math					
Katrina/Rita Share in Quartile 1	-0.63** (0.29)	-0.31 (0.36)	-0.92** (0.36)	-0.92* (0.50)	-0.65 (0.68)
Katrina/Rita Share in Quartile 2	0.28 (0.39)	-0.23 (0.57)	0.18 (0.49)	0.61 (0.49)	1.43* (0.84)
Katrina/Rita Share in Quartile 3	0.27 (0.35)	0.38 (0.62)	0.48 (0.44)	0.60 (0.50)	0.33 (0.63)
Katrina/Rita Share in Quartile 4	0.61 (0.49)	0.89 (0.76)	0.31 (0.60)	1.20* (0.63)	0.49 (0.87)
<i>Observations</i>	343,098	83,063	85,864	86,609	87,562
B. ELA					
Katrina/Rita Share in Quartile 1	0.72** (0.35)	0.77 (0.47)	0.91* (0.53)	0.51 (0.44)	1.10** (0.51)
Katrina/Rita Share in Quartile 2	0.40 (0.44)	0.69 (0.70)	0.33 (0.59)	0.36 (0.53)	-0.30 (0.48)
Katrina/Rita Share in Quartile 3	0.40 (0.45)	0.63 (0.74)	0.47 (0.57)	-0.09 (0.50)	-0.10 (0.55)
Katrina/Rita Share in Quartile 4	0.77 (0.47)	1.49 (0.94)	0.71 (0.69)	1.21** (0.55)	-0.85 (0.59)
<i>Observations</i>	343,081	83,230	85,956	86,318	87,577

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 38 - Nonlinear Models of Evacuee Share in School and Evacuee Achievement on Incumbent Achievement - Louisiana
Evacuee Share by Grade - Alterincumbent Standardization Using Only Pre-Katrina

	A. Elementary (No Lags)					B. Middle/High (Lags)				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Math										
Katrina/Rita Share in Quartile 1	-1.02 (0.64)	-2.05** (0.82)	-1.40** (0.64)	-2.30** (0.93)	-0.19 (1.35)	-1.40*** (0.34)	-0.82** (0.36)	-2.06*** (0.40)	-1.88*** (0.50)	-2.24*** (0.74)
Katrina/Rita Share in Quartile 2	-2.43*** (0.93)	-0.73 (1.16)	-0.56 (1.15)	-1.87 (1.36)	-4.86** (2.01)	-0.51 (0.42)	-0.50 (0.53)	-0.39 (0.50)	0.05 (0.50)	-0.20 (0.80)
Katrina/Rita Share in Quartile 3	0.77 (1.03)	-0.27 (1.31)	-0.49 (1.14)	-0.18 (1.13)	-0.15 (1.67)	0.93** (0.44)	0.24 (0.57)	1.36*** (0.51)	1.06** (0.53)	1.29* (0.67)
Katrina/Rita Share in Quartile 4	5.60*** (1.12)	4.06*** (1.55)	3.66*** (1.12)	6.61*** (1.23)	6.70*** (1.48)	1.21** (0.49)	1.99*** (0.69)	1.33** (0.52)	1.04* (0.55)	1.36* (0.74)
Observations	287,181	65,019	71,695	74,014	76,453	536,003	122,390	133,949	138,050	141,614
ELA										
Katrina/Rita Share in Quartile 1	0.07 (0.53)	-1.20 (0.88)	0.07 (0.59)	-1.15* (0.68)	-0.35 (0.74)	-0.57 (0.37)	-0.47 (0.42)	-0.38 (0.47)	-1.16*** (0.41)	-0.94* (0.54)
Katrina/Rita Share in Quartile 2	-1.25 (0.90)	-1.24 (1.38)	-1.21 (0.93)	-1.63* (0.97)	-1.76 (1.56)	-0.50 (0.44)	0.31 (0.64)	-0.39 (0.56)	-0.66 (0.52)	-0.41 (0.48)
Katrina/Rita Share in Quartile 3	-0.35 (0.64)	-0.41 (1.22)	-0.46 (0.80)	-0.51 (0.78)	-0.71 (0.89)	0.15 (0.47)	0.47 (0.65)	-0.30 (0.61)	0.02 (0.50)	0.07 (0.59)
Katrina/Rita Share in Quartile 4	1.14* (0.60)	1.31 (1.57)	0.52 (0.72)	1.52* (0.83)	1.65** (0.74)	1.46*** (0.39)	2.17*** (0.72)	1.57*** (0.53)	2.21*** (0.42)	0.96** (0.48)
Observations	287,230	65,243	72,005	73,791	76,191	522,387	116,426	128,315	136,122	141,524

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with the number of years between exams. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 39 - Nonlinear Models of Evacuee Share in Grade and Evacuee Achievement on Incumbent Achievement - Louisiana
Evacuee Share by Grade - Limiting to Schools with Positive Evacuee Share in 2005-06

	A. Elementary (No Lags)					B. Middle/High (Lags)				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Math										
Katrina/Rita Share in Quartile 1	0.37 (0.56)	-1.75** (0.72)	-0.94* (0.57)	-1.82** (0.80)	0.19 (1.13)	-0.79*** (0.25)	-0.78** (0.32)	-1.53*** (0.30)	-1.27*** (0.39)	-1.59*** (0.55)
Katrina/Rita Share in Quartile 2	-1.89** (0.74)	-0.90 (1.01)	-0.59 (0.93)	-1.35 (1.09)	-3.76** (1.57)	-0.17 (0.32)	-0.42 (0.45)	-0.34 (0.40)	-0.06 (0.39)	0.02 (0.62)
Katrina/Rita Share in Quartile 3	0.52 (0.89)	-0.23 (1.17)	-0.47 (1.00)	-0.34 (0.96)	-0.16 (1.33)	0.32 (0.32)	0.19 (0.47)	0.83** (0.42)	0.78* (0.40)	0.41 (0.52)
Katrina/Rita Share in Quartile 4	3.24*** (0.95)	3.02** (1.37)	2.83*** (0.97)	5.53*** (1.05)	5.25*** (1.22)	0.47 (0.32)	1.36** (0.55)	1.23*** (0.41)	0.73* (0.40)	0.97* (0.56)
Observations	242,501	55,069	60,266	62,498	64,668	495,142	124,125	129,409	127,333	129,409
ELA										
Katrina/Rita Share in Quartile 1	0.16 (0.57)	-1.38 (0.94)	0.13 (0.63)	-1.21* (0.73)	-0.40 (0.78)	-0.06 (0.29)	-0.33 (0.36)	-0.31 (0.39)	-0.88** (0.35)	-0.57 (0.44)
Katrina/Rita Share in Quartile 2	-1.57 (0.98)	-1.65 (1.46)	-1.13 (0.99)	-2.29** (1.02)	-2.17 (1.65)	0.03 (0.36)	0.65 (0.59)	-0.16 (0.48)	-0.28 (0.45)	-0.40 (0.43)
Katrina/Rita Share in Quartile 3	-0.34 (0.70)	-0.47 (1.29)	-0.63 (0.83)	-0.37 (0.82)	-0.68 (0.96)	-0.23 (0.36)	0.27 (0.60)	-0.09 (0.48)	-0.12 (0.43)	-0.08 (0.48)
Katrina/Rita Share in Quartile 4	0.79 (0.63)	1.23 (1.66)	0.33 (0.75)	1.52* (0.86)	1.66** (0.75)	0.39 (0.28)	1.59** (0.67)	1.18*** (0.45)	1.82*** (0.37)	0.52 (0.40)
Observations	242,539	55,037	60,595	62,299	64,608	482,306	107,536	119,528	125,670	129,572

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with the number of years between exams. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 40 - Estimates of Evacuee Share of Enrollment in School on
Incumbent Attendance and Disciplinary Infractions - Houston

	All	African- American	Boys	Girls	Placebo Test
	(1)	(2)	(3)	(4)	(5)
A. Attendance Rates					
i. Elementary					
Attendance Rate	-0.61 (0.78)	-2.21 (1.79)	-1.04 (0.93)	-0.09 (0.93)	-1.02 (0.70)
Observations	223,776	60,710	115,104	108,672	123,366
ii. Middle/High					
Attendance Rate	-5.0 (3.9)	-9.2*** (3.2)	-6.0 (4.8)	-3.9 (3.1)	-5.6 (3.5)
Observations	323,499	98,595	163,133	160,366	166,202
B. Disciplinary Infractions					
i. Elementary					
Disciplinary Infractions	0.04 (0.20)	-0.17 (0.47)	0.12 (0.31)	-0.05 (0.14)	-0.23 (0.36)
Observations	223,776	60,710	115,104	108,672	123,366
ii. Middle/High					
Disciplinary Infractions	2.29 (1.46)	3.88** (1.79)	2.27 (1.85)	2.19** (1.10)	-1.11 (1.60)
Observations	323,499	98,595	163,133	160,366	166,202

Standard errors are provided in parentheses and clustered by school. Disciplinary infractions in Houston is the number of times a student receives an in-school suspension or more severe punishment in a year. Regressions cover 2003-04 - 2006-07 and include student's race, gender, free lunch/reduced price lunch status, school fixed-effects, grade-by-year fixed-effects, and a pre-Katrina lagged dependent variable. The placebo test in column (7) include 2003-04 and 2004-05 only and apply 2005-06 Katrina/Rita share to 2004-05 observations. Elementary is defined as any student in grades 1 - 5. Middle/High is any student in grade 6 - 12. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 41 - Behavior Specification Tests
Houston, Grade Level Evacuee Share

		A. Elementary		B. Middle/High	
		Attendance	Discipline	Attendance	Discipline
(1)	IV (Shelter Instrument Only)	0.73 (1.87)	0.29 (0.48)	0.5 (6.6)	3.8 (3.3)
(2)	IV (Shelter & 9/13/05 Instruments)	0.35 (1.17)	0.28 (0.34)	-2.1 (5.9)	2.8 (2.7)
(3)	Student & School Fixed Effects	0.49 (0.81)	-0.09 (0.17)	-6.9* (3.7)	0.50 (0.97)
	Observations	223,776	223,776	323,499	323,499
(4)	Limited to Schools with < 10% Katrina in 0506	1.56* (0.80)	-0.03 (0.23)	-8.7* (4.4)	2.2 (1.8)
	Observations	219,371	219,371	304,126	304,126
(5)	Interrupted Panel - 2003-04, 05-06, and 06-07 Only	0.03 (0.80)	-0.16 (0.21)	-6.2 (4.1)	1.5 (1.5)
	Observations	162,180	162,180	239,086	239,086
	Instrument for Average Peer Outcome w/ Evac Share				
	OLS - All Years	0.26*** (0.08)	0.61*** (0.03)	0.51*** (0.08)	0.88*** (0.02)
(6)	Average Peer Outcome in School				
	First Stage	-6.77*** (0.85)	0.14 (0.22)	-20.0*** (2.9)	1.55 (1.10)
	Grade Level Katrina Share				
	Second Stage	-0.03 (0.11)	-0.37 (1.56)	0.20 (0.15)	1.20*** (0.38)
	Average Peer Outcome in School				
	Observations	223,764	223,764	323,477	323,477
(7)	OLS - Pre-Katrina Years	0.25*** (0.05)	0.54*** (0.05)	0.32*** (0.08)	0.82*** (0.03)
	Average Peer Outcome in School				
	Observations	123,955	123,955	166,551	166,551
	Instrument for Evac Share w/ Evac Share in 0405 School				
	OLS	0.29 (0.75)	-0.02 (0.17)	0.3 (3.2)	0.19 (1.20)
(8)	Grade Level Katrina Share				
	Second Stage	0.47 (0.89)	-0.07 (0.17)	4.7 (3.8)	-0.63 (1.29)
	Grade Level Katrina Share				
	Observations	181,628	181,628	224,370	224,370
(9)	School - Year Fixed Effects (2006, 2007 Only)	1.79* (1.08)	-0.04 (0.27)	0.8 (4.2)	0.03 (2.13)
	Observations	99,816	99,816	156,934	156,934
(10)	School - Grade Fixed Effects	0.10 (0.69)	-0.06 (0.15)	-3.8 (3.5)	2.03* (1.20)
	Observations	223,776	223,776	323,499	323,499
(11)	Evacuee Counts in Grade with Quartic Control in	-0.0024 (0.0052)	-0.0007 (0.0012)	-0.0040 (0.0052)	0.0023 (0.0027)
	Grade-Level Enrollment				
	Observations	223,776	223,776	323,499	323,499
	Separate Estimates for New Entrants to School				
(12)	Evacuee Share in Grade	0.73 (0.83)	-0.09 (0.16)	-1.3 (3.4)	0.69 (1.10)
	Evacuee Share in Grade X New Entrant	-2.77 (2.37)	0.17 (0.38)	-8.2* (4.3)	3.65** (1.78)
	Observations	223,776	223,776	323,499	323,499
(13)	Exclude Schools With No Evacuees in 2005-06	0.17 (0.74)	-0.09 (0.15)	-3.3 (3.5)	1.94 (1.22)
	Observations	196,543	196,543	314,033	314,033

Standard errors are provided in parentheses and clustered by school. Disciplinary infractions in Houston is the number of times a student receives an in-school suspension or more severe punishment in a year. Regressions cover 2003-04 - 2006-07 except where noted and include student's race, gender, free/reduced price lunch status, school fixed-effects, grade-by-year fixed effects, and a pre-Katrina lagged dependent variable. 2SLS regressions use Katrina/Rita share on 9/13/05 as an instrument for evacuee share. Elementary covers students in grades 1 - 5 prior to 2006, grades 2 - 5 in 2006, and grades 3 - 5 in 2007. Middle/High is any student in grades 6 - 12. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 42 - Effect of Disruption on Attendance and Discipline - Houston

	A. Attendance		B. Discipline	
	I. Elementary	II. Middle/High	I. Elementary	II. Middle/High
<i>i. OLS w/o Entry</i>				
Avg Weekly Katrina Share in Grade	-0.12 (0.83)	-9.61* (5.11)	0.02 (0.18)	3.69** (1.79)
<i>ii. OLS w/ Entry</i>				
Avg Weekly Katrina Share in Grade	0.10 (0.85)	-9.98* (5.15)	-0.08 (0.18)	3.74** (1.78)
Std Deviation of Weekly Relative Enrollment	-1.09 (1.11)	1.96 (1.89)	0.50*** (0.19)	-0.27 (0.33)
<i>iii. 2SLS - First Stage - (SD of Weekly Enrollment is Endogenous)</i>				
Avg Weekly Katrina Share in Grade	-0.00 (0.02)	-0.22 (0.18)	-0.00 (0.02)	-0.22 (0.18)
Std Deviation of Weekly Katrina Share	0.81*** (0.07)	1.58** (0.62)	0.81*** (0.07)	1.58** (0.62)
<i>iv. 2SLS - Second Stage</i>				
Avg Weekly Katrina Share in Grade	-0.11 (1.11)	-12.34 (7.50)	0.21 (0.33)	4.95** (2.17)
Std Deviation of Weekly Relative Enrollment	-0.04 (4.65)	14.28 (17.87)	-0.90 (1.33)	-6.57 (5.47)
<i>Observations</i>	223,776	323,499	223,776	323,499

Standard errors are provided in parentheses and clustered by Grade. Disciplinary infractions in Houston is the number of times a student receives an in-Grade suspension or more severe punishment in a year. RRegressions cover 2003-04 - 2006-07 except where noted and include student's race, gender, free/reduced price lunch status, school fixed-effects, grade-by-year fixed effects, and a pre-Katrina lagged dependent variable Elementary covers students in grades 1 - 5 prior to 2006, grades 2 - 5 in 2006, and grades 3 - 5 in 2007. Middle/High is any student in grades 6 - 12. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 43 - Nonlinear Models of Evacuee Grade in School and Evacuee Achievement & Discipline in Incumbent Achievement - Louisiana

	A. Elementary (No Lags)					B. Middle/High (with Lags)				
	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
Pre-Katrina LEAP Quartile	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Math										
Katrina/Rita Share in Quartile 1	0.50 (0.67)	-1.00 (0.81)	-0.59 (0.64)	-1.44 (0.89)	0.42 (1.16)	-0.73*** (0.25)	-0.64** (0.31)	-1.50*** (0.29)	-1.20*** (0.38)	-1.60*** (0.54)
Katrina/Rita Share in Quartile 2	-1.78** (0.72)	-0.44 (0.98)	-0.43 (0.92)	-1.20 (1.10)	-3.72** (1.57)	-0.16 (0.32)	-0.54 (0.45)	-0.34 (0.40)	-0.04 (0.40)	0.15 (0.64)
Katrina/Rita Share in Quartile 3	0.55 (0.89)	0.12 (1.15)	-0.31 (1.02)	-0.19 (0.97)	-0.13 (1.33)	0.40 (0.31)	0.39 (0.48)	0.74* (0.41)	0.81** (0.40)	0.48 (0.51)
Katrina/Rita Share in Quartile 4	3.33*** (0.96)	3.25** (1.32)	2.81*** (0.96)	5.50*** (1.05)	5.17*** (1.22)	0.46 (0.32)	1.22** (0.55)	1.25*** (0.41)	0.87** (0.42)	1.05* (0.60)
Katrina/Rita Share w/ Disciplinary Infraction in 2004-05	-0.04 (1.05)	-1.60 (1.24)	-1.22 (1.11)	-1.41 (1.29)	-1.25 (1.79)	-0.15 (0.09)	-0.10 (0.15)	-0.13 (0.10)	-0.22 (0.16)	-0.27** (0.13)
Observations	287,181	65,025	71,606	74,138	76,412	531,055	122,734	132,919	136,480	138,922
ELA										
Katrina/Rita Share in Quartile 1	0.33 (0.64)	-0.18 (1.04)	0.38 (0.79)	-1.06 (0.84)	0.43 (0.78)	-0.11 (0.28)	-0.41 (0.35)	-0.41 (0.39)	-0.96*** (0.35)	-0.42 (0.44)
Katrina/Rita Share in Quartile 2	-1.33 (0.96)	-1.35 (1.47)	-1.08 (0.98)	-2.09** (1.01)	-2.09 (1.62)	0.08 (0.35)	0.80 (0.58)	-0.09 (0.48)	-0.36 (0.45)	-0.50 (0.42)
Katrina/Rita Share in Quartile 3	-0.25 (0.70)	-0.10 (1.25)	-0.56 (0.83)	-0.29 (0.80)	-0.18 (0.94)	-0.19 (0.35)	0.22 (0.58)	0.04 (0.49)	-0.12 (0.42)	0.01 (0.48)
Katrina/Rita Share in Quartile 4	0.84 (0.64)	1.62 (1.49)	0.39 (0.76)	1.53* (0.86)	1.78** (0.84)	0.38 (0.27)	1.64** (0.66)	1.10** (0.45)	1.81*** (0.37)	0.56 (0.42)
Katrina/Rita Share w/ Disciplinary Infraction in 2004-05	0.01 (0.92)	-3.03** (1.47)	-0.68 (1.45)	0.06 (1.17)	-2.89** (1.20)	-0.12 (0.08)	-0.12 (0.17)	-0.20 (0.13)	-0.10 (0.08)	-0.12 (0.11)
Observations	287,230	65,210	72,013	73,885	76,122	517,438	115,779	128,069	134,615	138,975

Standard errors are provided in parentheses and clustered by school. LEAP scores are standard deviations of scale scores within grade and year. The standardization uses non-evacuees in schools where there are < 70% evacuees and excludes schools in Orleans, Jefferson, St. Bernard, Plaquemines, Cameron, and Calcasieu parishes. When students have multiple scores for a single subject in a given year we use the lowest score. Quartiles are determined using the most recent pre-Katrina score available and are within grade and year. Disciplinary infractions in LA are the number of times in a year the student was given a suspension or expulsion and data is limited to students who took a LEAP or iLEAP exam. Regressions also include controls for gender, race, free/reduced-price lunch eligibility, school fixed-effects, grade-by-year fixed-effects, and, for middle/high, lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. Elementary includes grade 4 from 1999-2000 through 2004-2005 and grade 5 in 2005-2006. Middle/High includes grade 10 in 2001-2002 & 2002-2003, grades 8 and 10 in 2003-2004 & 2004-2005, and grades 6 - 10 in 2005-2006 & 2006-2007. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 44 - Effect of Evacuee Share in School on Class Sizes
by Evacuee Avg Math & Reading Achievement Levels - LA

	% of Elementary Classes of Size 1 to 20	% of Elementary Classes of Size 21 - 26	% of Elementary Classes of Size 27+
Katrina/Rita Share in Quartile 1	0.55 (0.55)	-0.87* (0.52)	0.32 (0.24)
Katrina/Rita Share in Quartile 2	-0.43 (0.99)	0.37 (0.96)	0.06 (0.48)
Katrina/Rita Share in Quartile 3	-0.04 (0.63)	0.21 (0.57)	-0.17 (0.31)
Katrina/Rita Share in Quartile 4	0.26 (0.52)	-0.01 (0.42)	-0.25 (0.28)
<i>Observations</i>	2,754	2,754	2,754
	% of Mid/High Classes of Size 1 to 20	% of Mid/High Classes of Size 21 - 26	% of Mid/High Classes of Size 27+
Katrina/Rita Share in Quartile 1	-0.14 (0.65)	0.42 (0.45)	-0.20 (0.46)
Katrina/Rita Share in Quartile 2	-0.18 (0.91)	0.94 (0.74)	-0.55 (0.65)
Katrina/Rita Share in Quartile 3	1.59 (1.52)	-0.60 (0.78)	-0.06 (0.84)
Katrina/Rita Share in Quartile 4	-1.19 (1.18)	0.09 (0.48)	0.18 (0.56)
<i>Observations</i>	2,309	2,309	2,309

Unit of observation is the school-year. Regressions cover 2002-03 through 2006-07 and include school fixed-effects, year dummies, % of tested students who are African-American, hispanic, asian, free/reduced-price lunch, and the distribution of tested students across grades in each year. Elementary is defined as any school covering at least one of grades KG - 5. Middle/High is any covering at least one of grades 6 - 12. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 45 Effect of Evacuee Share in School on School Resources and Staffing in Houston

A. Elementary								
	Operating Expenditures	Instructional Expenditures	Avg Teacher Experience	% of Teachers Beginning	% of Teachers 1 - 5 Yrs Exp	% of Teachers w/ Grad Degree	% of Teachers w/ Certification	Avg Class Size (Grades KG - 5)
Katrina/Rita Fraction	-1,787 (5,087)	342 (3,856)	-3.02 (5.86)	22.81 (19.48)	2.90 (25.46)	0.337** (0.169)	0.313* (0.168)	-3.25 (8.97)
<i>Observations</i>	<i>791</i>	<i>791</i>	<i>763</i>	<i>766</i>	<i>766</i>	<i>763</i>	<i>763</i>	<i>791</i>
B. Middle/High								
	Operating Expenditures	Instructional Expenditures	Avg Teacher Experience	% of Teachers Beginning	% of Teachers 1 - 5 Yrs Exp	% of Teachers w/ Grad Degree	% of Teachers w/ Certification	Avg Class Size Math & Reading
Katrina/Rita Fraction	12,748 (13,717)	8,411 (9,764)	-1.66 (13.00)	6.40 (32.39)	3.53 (53.07)	-0.567 (0.345)	0.602 (0.561)	32.74 (27.28)
<i>Observations</i>	<i>492</i>	<i>492</i>	<i>464</i>	<i>471</i>	<i>471</i>	<i>464</i>	<i>464</i>	<i>325</i>

Unit of observation is the school-year. All expenditures are per-student. Regressions cover 2003-04 - 2006-07 and include school fixed-effects and year dummies, % of school at each grade level as well as % African-American, Hispanic, asian, Incumbent American, and economically disadvantaged. Elementary is defined as any school covering at least one of grades KG - 5. Middle/High is any covering at least one of grades 6 - 12. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table 46 School Switching in Response to Evacuee Shares in Grade
by Pre-Katrina Achievement Quartile - Houston

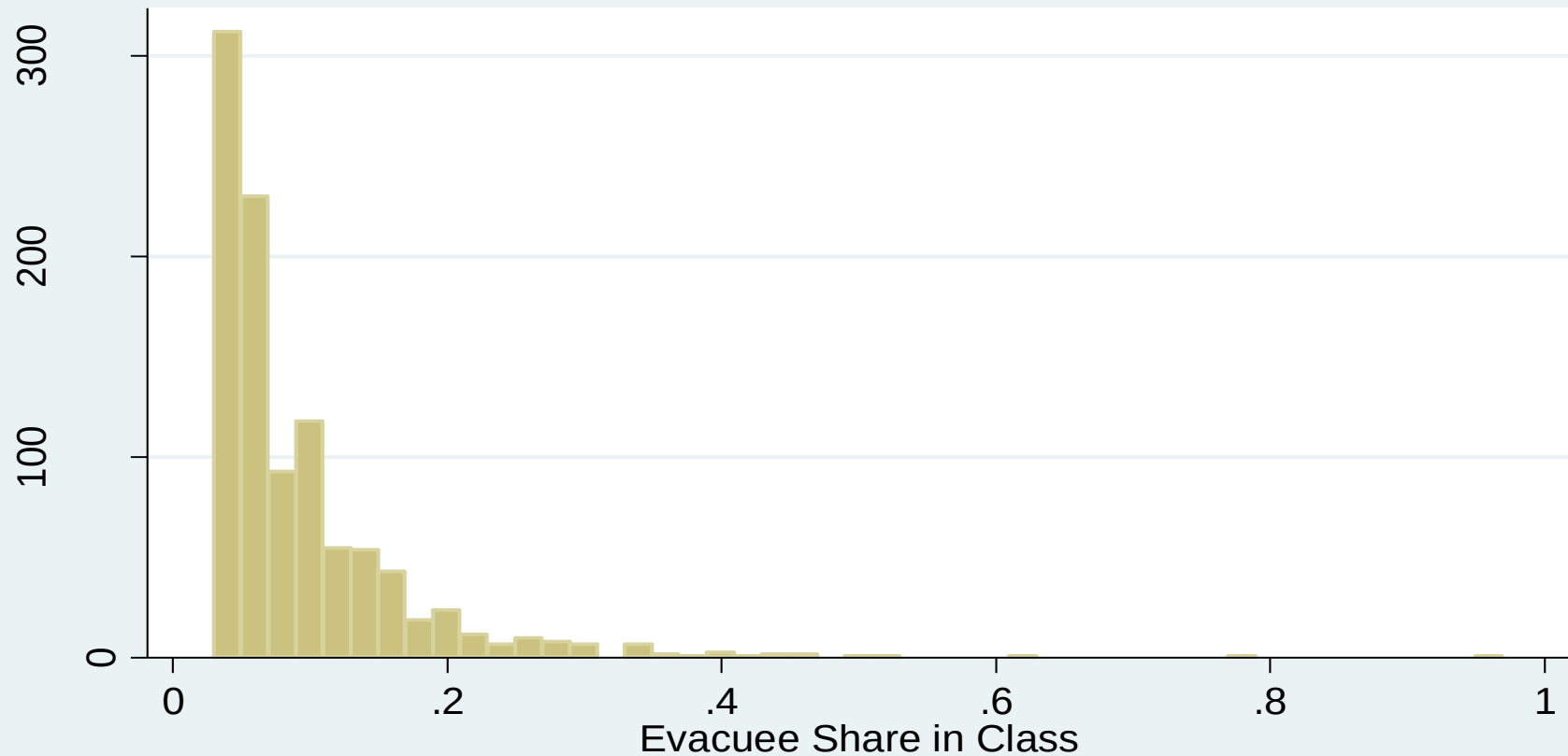
A. Elementary					
	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
	(1)	(2)	(3)	(4)	(5)
<i>i. Student switches between HISD schools in year $t + 1$[†]</i>					
Katrina/Rita	0.110	0.216	0.197	0.474	-0.256
Fraction	(0.153)	(0.357)	(0.312)	(0.420)	(0.300)
Observations	283,628	26,277	26,040	26,003	25,624
Pre-Katrina Mean	0.122	0.178	0.147	0.123	0.094
<i>ii. Student leaves HISD in year $t + 1$^{† ‡}</i>					
Katrina/Rita	-0.085*	0.108	0.136	0.266*	0.376**
Fraction	(0.046)	(0.189)	(0.215)	(0.160)	(0.157)
Observations	340,963	38,412	38,430	38,431	38,420
Pre-Katrina Mean	0.153	0.093	0.095	0.098	0.095
B. Middle/High					
	All	Bottom Quartile	2nd Quartile	3rd Quartile	Top Quartile
	(1)	(2)	(3)	(4)	(5)
<i>i. Student switches between HISD schools in year $t + 1$[†]</i>					
Katrina/Rita	0.051	0.281**	0.084	-0.011	-0.095
Fraction	(0.065)	(0.127)	(0.111)	(0.092)	(0.070)
Observations	275,212	57,681	57,323	57,114	57,515
Pre-Katrina Mean	0.088	0.133	0.111	0.081	0.050
<i>ii. Student leaves HISD in year $t + 1$^{† ‡}</i>					
Katrina/Rita	-0.014	-0.047	0.065	0.111	0.049
Fraction	(0.087)	(0.173)	(0.113)	(0.105)	(0.105)
Observations	331,456	69,952	70,011	70,025	70,009
Pre-Katrina Mean	0.179	0.153	0.103	0.087	0.0639

[†] Limited to students not in the maximum grade for a school in year t . [‡] Excludes grade 12.

Standard errors are provided in parentheses and clustered by school. Regressions cover 2003-04 - 2006-07 and include student's race, gender, free/reduced price lunch status, and school fixed-effects. Quartiles defined by average of students' most-recent pre-Katrina math and reading scores. Elementary is defined as any student in grades 1 - 5. Middle/High is any student in grade 6 - 12. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.

Appendix Figure 1: Distribution of Houston Evacuee Shares Across Classrooms in Grades 1 to 5 - Houston, 2005-06

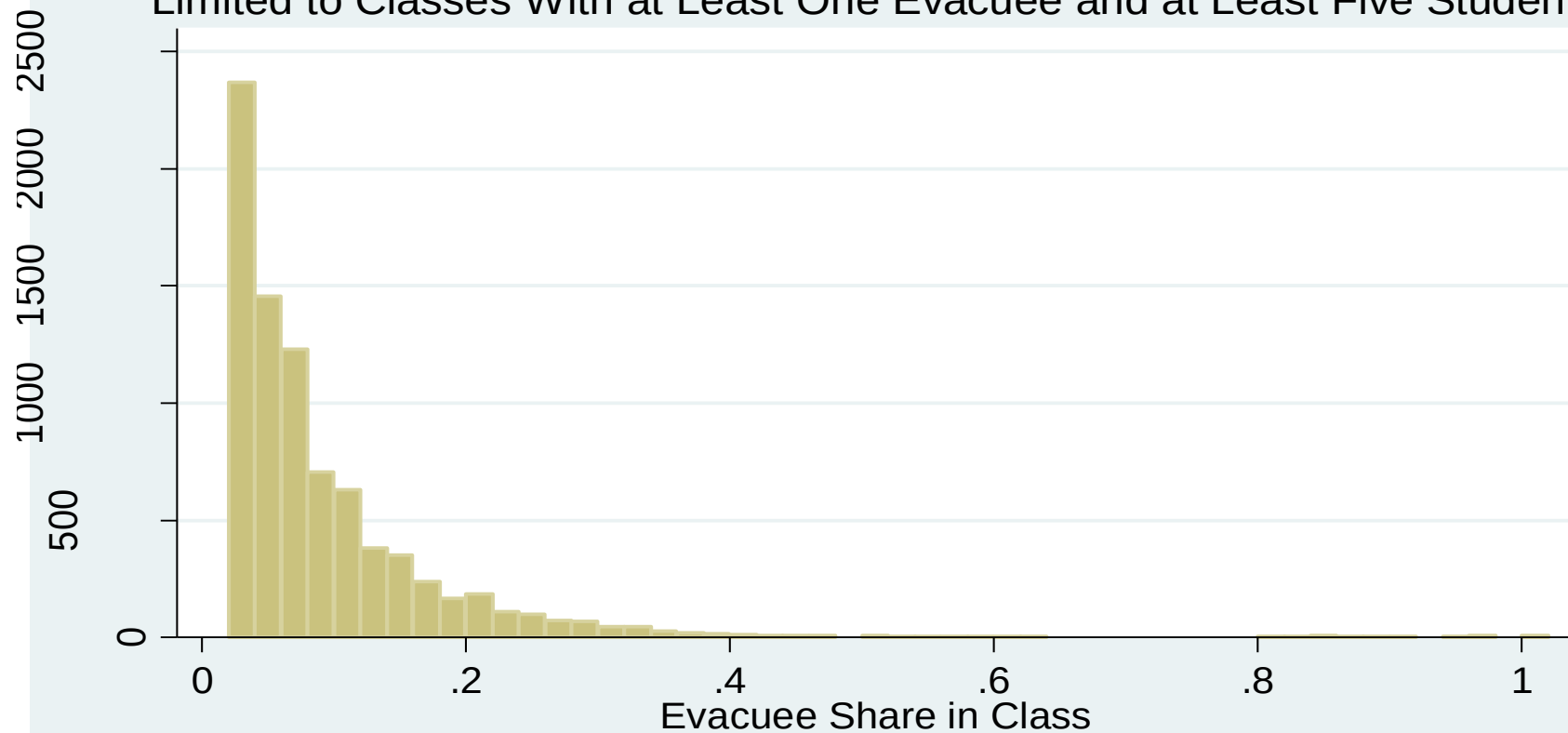
Limited to Classes With at Least One Evacuee and at Least Five Students



17% of classes of size ≥ 5 across HISD have one or more evacuees. Classes are determined by teacher-grade-school combinations at end of year.

Appendix Figure 2: Distribution of Houston Evacuee Shares Across Classrooms in Grades 6 to 12 - Houston, 2005-06

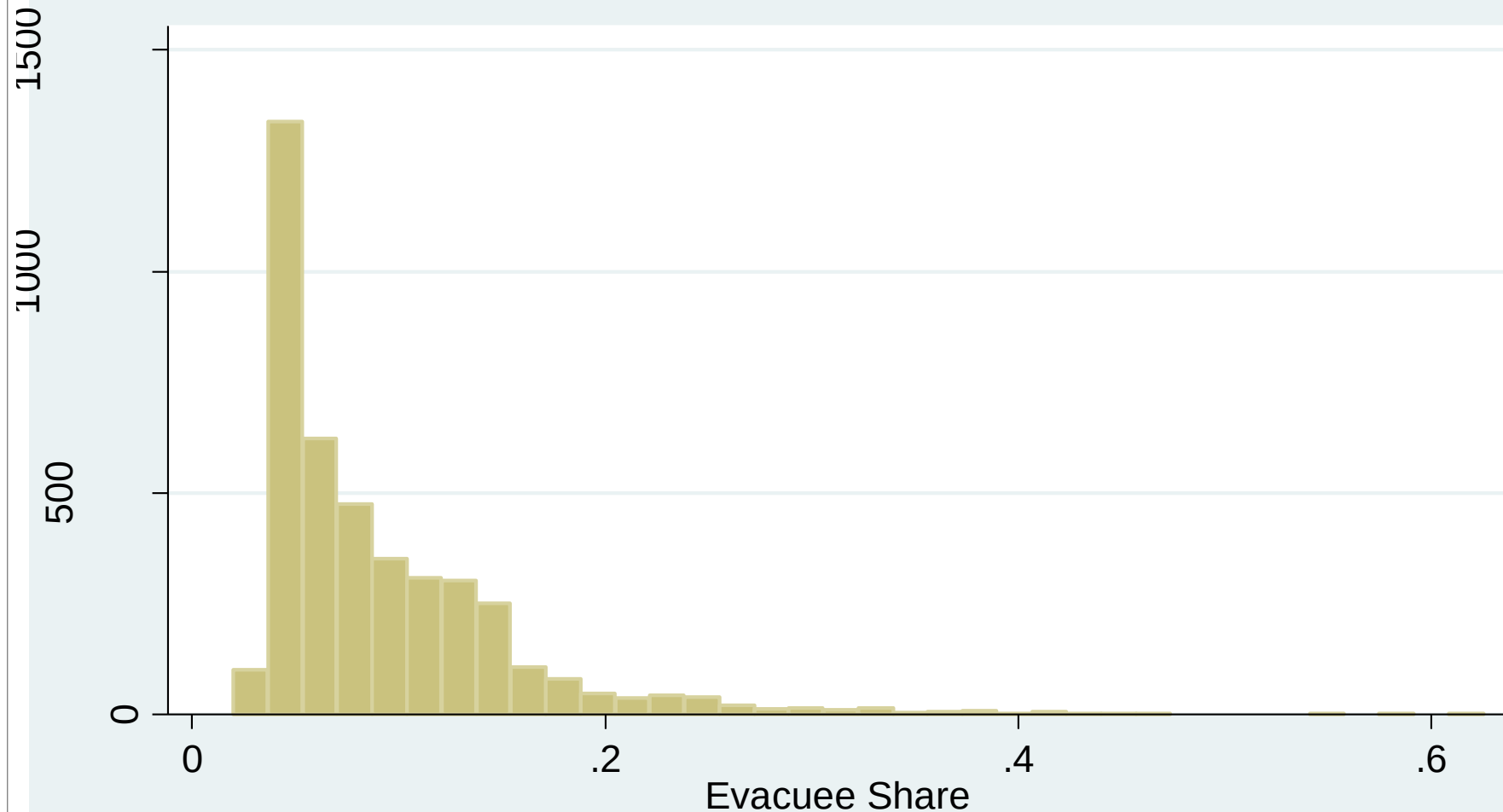
Limited to Classes With at Least One Evacuee and at Least Five Students



32% of classes of size ≥ 5 across HISD have one or more evacuees. Classes are for fall semester, spring semester, or year-round. Students are identified as being in class if they receive a final grade for that class.

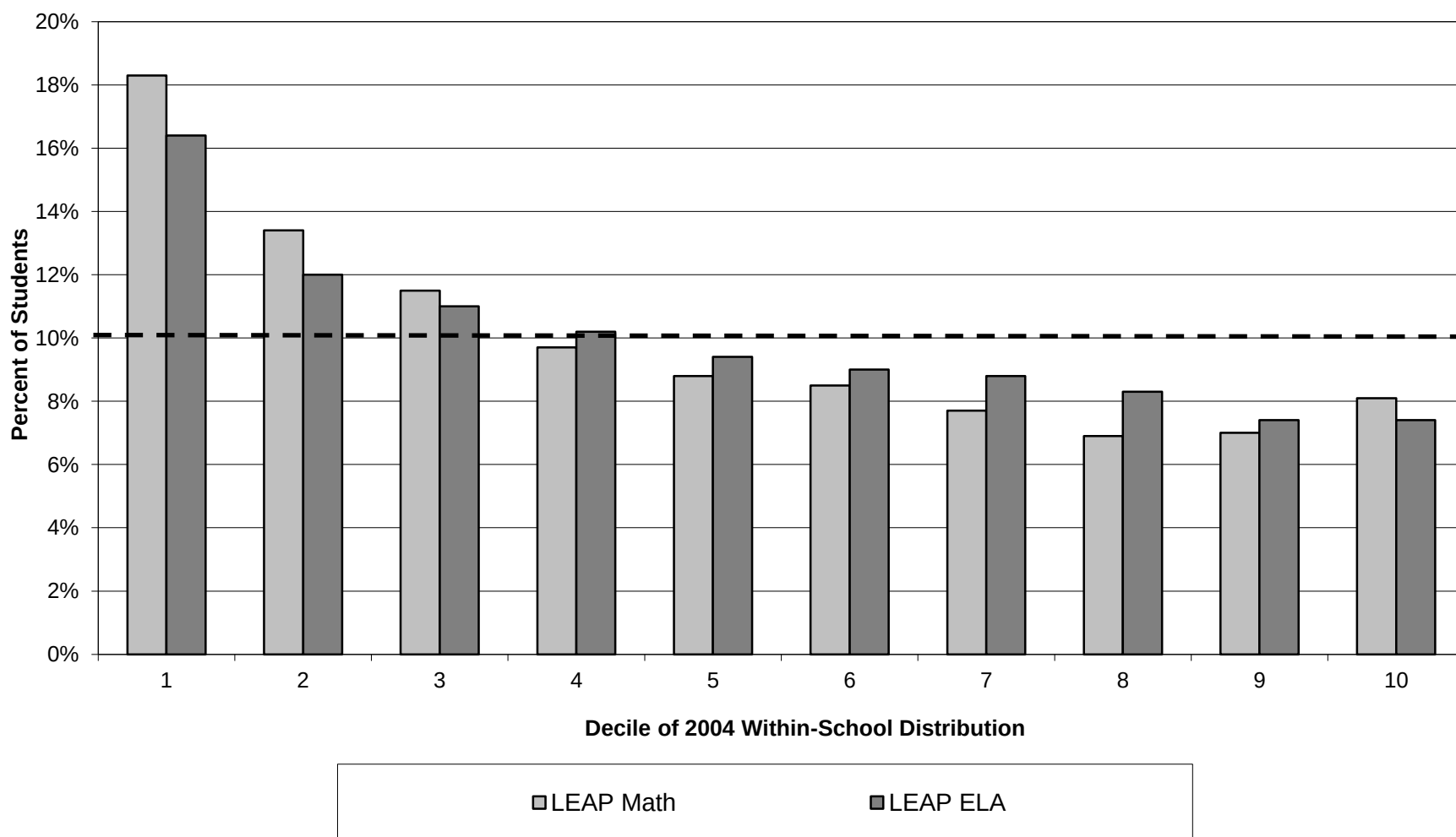
Appendix Figure 3: Distribution of Louisiana Evacuee Shares Across Test Administrators, 2005-06

Limited to Test Administrators With at Least One Evacuee and Five Students



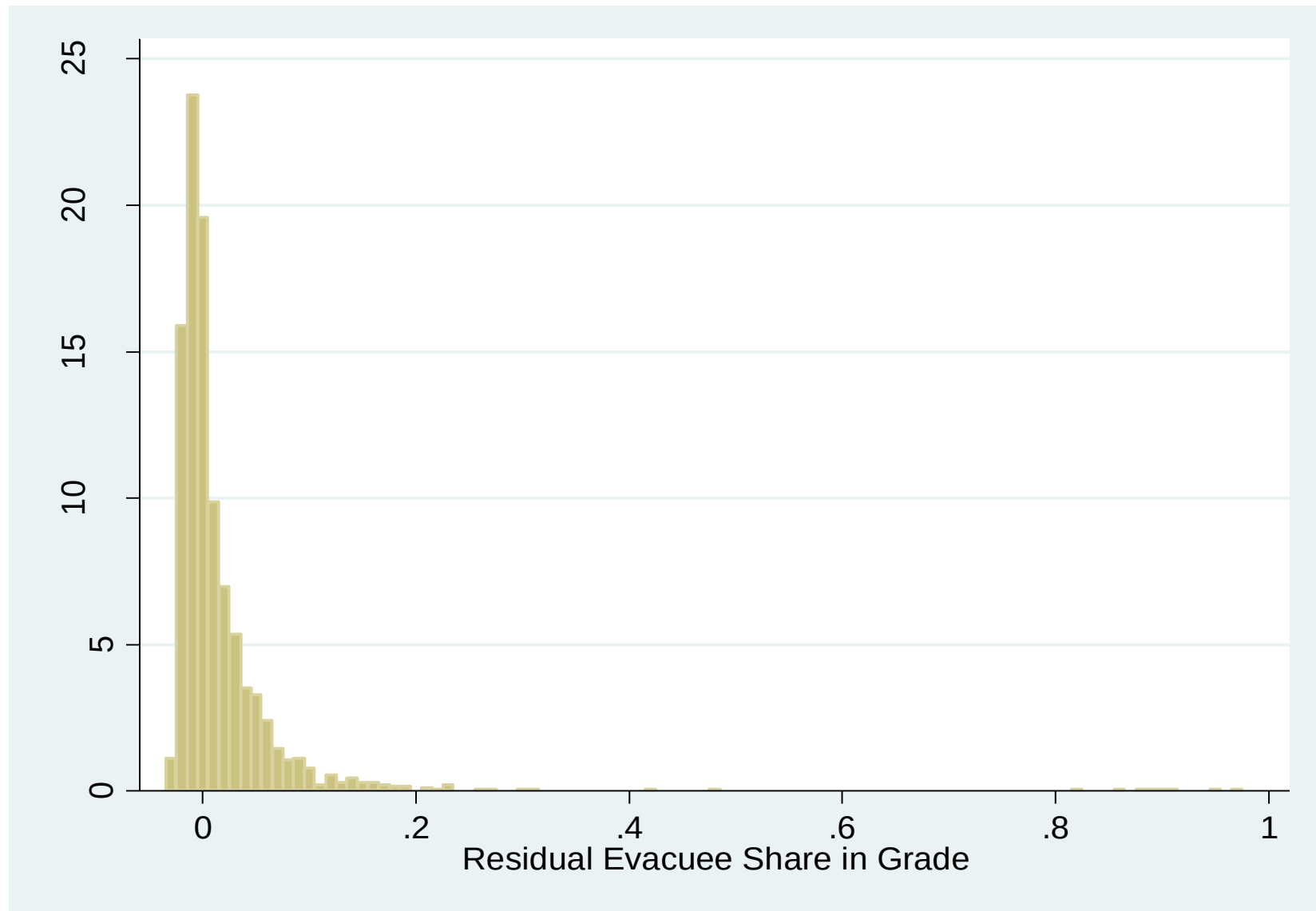
35% of test administrators with at least 5 students are linked to at least one evacuee.

Appendix Figure 4: Distribution of Evacuees's Lagged Test Scores in Same School Incumbent Distribution - Louisiana, 2005-06

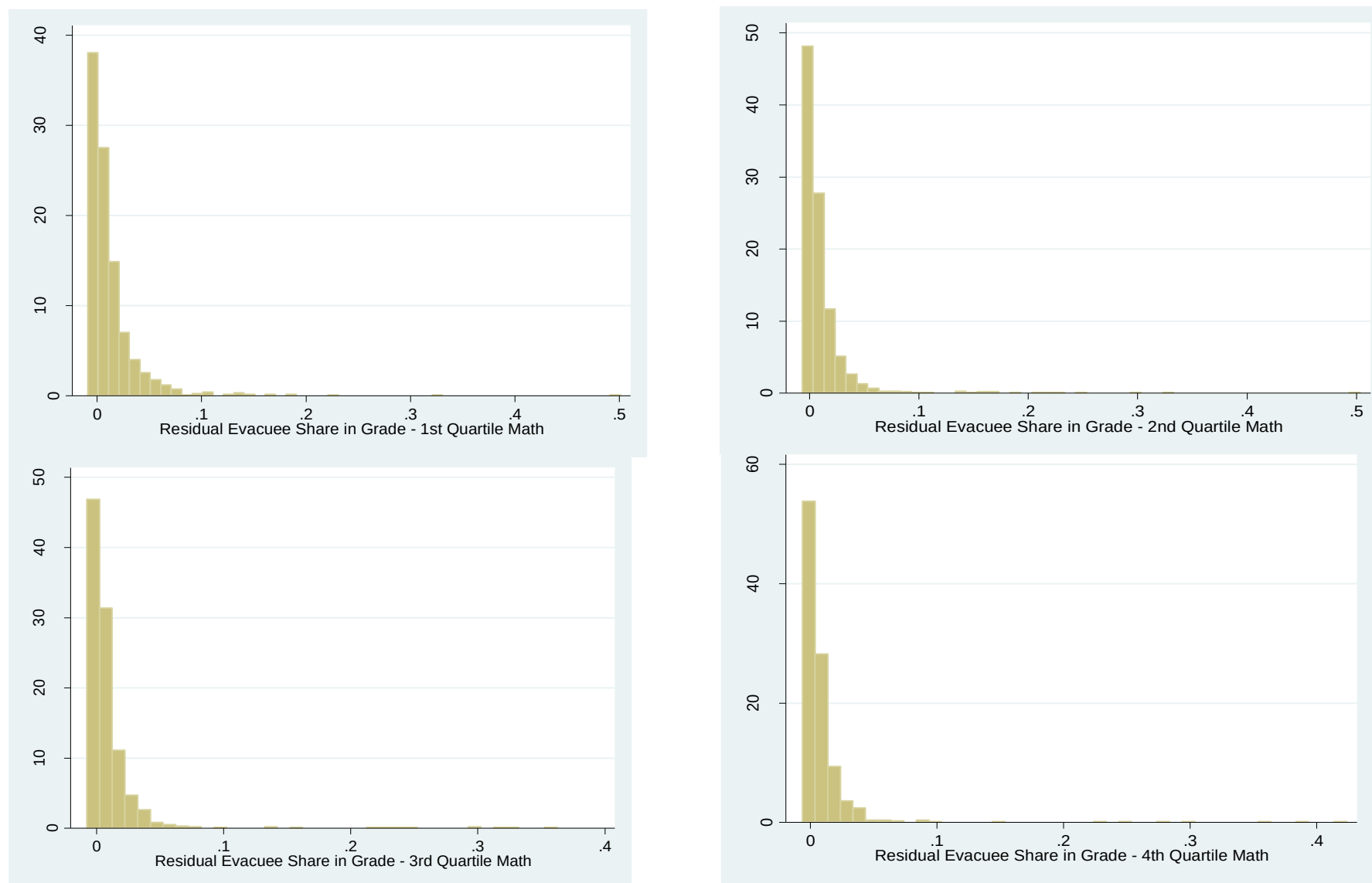


Each bar shows the percent of evacuee students in 2005-06 whose lagged test scores are in the listed decile of the within-school incumbent student lagged achievement distribution.

Appendix Figure 5 - Distribution of Residuals from School-Grade Level Regressions of 2005-06 Evacuee Shares on Mean Lagged Math and ELA Scores, Conditional on Having a Positive Evacuee Share in the Given Quartile - LA



Appendix Figure 6 - Distribution of Residuals from School-Grade Level Regressions of 2005-06 Evacuee Shares by Evacuees' Math Achievement Quartiles on Mean Lagged Math and ELA Scores
Conditional on Having a Positive Evacuee Share in the Given Quartile - LA



Appendix Figure 7 - Distribution of Residuals from School-Grade Level Regressions of 2005-06 Evacuee Shares by Evacuees' ELA Achievement
Quartiles on Mean Lagged Math and ELA Scores
Conditional on Having a Positive Evacuee Share in the Given Quartile - LA

